# TRIBHUVAN UNIVERSITY — FACULTY OF MANAGEMENT

## BBA / BBM Eighth Semester — April 2026 Examination Papers Complete Solutions
## Includes All Questions, Theory, Step-by-Step Numericals, Diagrams & Exam Memory Hints

**Subjects Covered in This Combined Solution PDF (All 4 Papers from Attached Past Question PDF):**
1. **MGT 240: Strategic Management** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)
2. **BNK 221: Corporate Financing Decision** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)
3. **BNK 202: Financial Derivatives** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)
4. **BNK 209: Insurance and Risk Management** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)

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# PAPER 1: TRIBHUVAN UNIVERSITY — MGT 240: STRATEGIC MANAGEMENT

**BBA / BBM / Eighth Semester | April 2026 | Full Marks: 100 | Pass Marks: 50 | Time: 3 Hrs**

## Group 'A': Brief Answer Questions [10 x 2 = 20]

### Q1. What is meant by organization's vision?

**Answer:** An organization's vision (or strategic vision) is a forward-looking declaration of what the organization aspires to become and achieve in the long-term future (typically 5 to 20 years). It provides a mental picture of the desired future state, inspires stakeholders, and guides strategic direction.

### Q2. Mention features of strategic decision.

**Answer:** The major features of a strategic decision are:
• **1. Long-Term & Future-Oriented:** Determines the direction and survival of the firm over many years.
• **2. Top-Management Involvement:** Formulated by the Board of Directors and CEO with organization-wide scope.
• **3. Substantial Resource Commitment:** Requires major financial, physical, and human resource allocation and is costly to reverse.
• **4. Environment-Matching:** Aligns internal capabilities with external environmental opportunities and threats.

### Q3. Write two strategic directions.

**Answer:** Based on Igor Ansoff's Product-Market Matrix, two primary strategic directions for growth are:
• **1. Market Penetration / Market Development:** Increasing market share with existing products in current markets (Penetration) or entering new geographical/demographic markets (Market Development).
• **2. Product Development / Diversification:** Introducing new or modified products to existing markets (Product Development) or entering entirely new products and new markets (Diversification).

### Q4. Differentiate between threshold resources and strategic resources.

**Answer:**
• **Threshold Resources:** The minimum baseline tangible and intangible resources needed for an organization simply to exist, enter an industry, and meet basic customer requirements (they ensure survival/parity but do not give a competitive advantage).
• **Strategic (Unique) Resources:** Rare, valuable, inimitable, and non-substitutable (VRIO) resources owned by a firm that underpin its core competencies and create a sustainable competitive advantage over rivals.

### Q5. Define scenario planning.

**Answer:** Scenario planning is a structured strategic forecasting technique in which strategists construct several plausible, internally consistent alternative views ("scenarios"—such as optimistic, pessimistic, and most-likely futures) of how key external environmental drivers (political, economic, technological) might evolve, enabling the firm to prepare flexible contingency strategies.

### Q6. Mention the resources that create competitive advantages.

**Answer:** According to Barney's Resource-Based View (RBV / VRIO Framework), resources that create sustainable competitive advantage must possess four attributes:
• **1. Valuable (V):** Create customer value or exploit opportunities / neutralize threats.
• **2. Rare (R):** Possessed by only one or very few competing firms.
• **3. Inimitable (I):** Costly or impossible for competitors to copy or imitate (due to patents, brand heritage, or causal

ambiguity).
• **4. Organized / Non-Substitutable (O/N):** Supported by organizational systems and having no strategic equivalent.

### Q7. Sketch SBU structure with suitable labeling.

**Answer:** A Strategic Business Unit (SBU) organizational structure groups similar divisions/product lines into autonomous business units, each headed by a VP/General Manager accountable for its own strategy, sales, and profits under Corporate HQ:




### Q8. Give four characteristics of strategic evaluation.

**Answer:** Four essential characteristics of effective strategic evaluation are:
• **1. Economical & Meaningful:** Should use minimum necessary information without overburdening managers.
• **2. Timely & Action-Oriented:** Must detect deviations promptly so corrective action can be taken before damage occurs.
• **3. True Picture of Performance:** Should measure both quantitative financial results and qualitative long-term strategic health.
• **4. Operational Flexibility:** Should adapt to changing environmental conditions rather than enforcing rigid controls.

### Q9. Make a list of operational strategy for HR department.

**Answer:** Operational (functional) strategies for the Human Resource (HR) department include:
• **1. Talent Acquisition & Recruitment Strategy:** Hiring right-skilled employees aligned with corporate strategy.
• **2. Training, Learning & Development Strategy:** Continuous upskilling and leadership development programs.
• **3. Performance Appraisal & Reward Strategy:** Linking compensation, incentives, and promotions to strategic KPIs.
• **4. Employee Retention & Quality of Work-Life Strategy:** Reducing turnover through career growth, safety, and employee engagement.

### Q10. Suggest four ways to reduce cost of production.

**Answer:** Four effective ways to reduce the cost of production are:
• **1. Economies of Scale & Experience Curve:** Producing in larger volumes to spread fixed overhead costs and leverage worker learning curves.
• **2. Process Automation & Modern Technology:** Adopting lean manufacturing, CAD/CAM, and automation to minimize labor hours and scrap/wastage.
• **3. Supply Chain Optimization & Just-in-Time (JIT) Inventory:** Negotiating bulk supplier discounts and using JIT to eliminate warehousing and holding costs.
• **4. Value Engineering & Product Redesign:** Simplifying product design and standardizing parts across product lines.

## Group 'B': Short Answer Questions (Attempt any SIX — ALL Solved) [6 x 5 = 30]

### Q11. Describe the process of strategic management.

**Answer:** The strategic management process is a continuous, dynamic sequence of four core phases through which an organization defines and achieves its long-term mission and objectives:
• **1. Environmental Scanning & Strategic Analysis (Where are we now?):**
- Setting/reviewing the Vision, Mission, and Objectives.
- Conducting External Analysis (PESTEL macro-environment and Porter's Five Forces industry analysis) to identify Opportunities and Threats (OT) -> summarized in an ETOP / EFE matrix.
- Conducting Internal Analysis (Resource-Based View, Value Chain, functional audit) to identify Strengths and Weaknesses (SW) -> summarized in a SAP / IFE matrix.
• **2. Strategy Formulation & Strategic Choice (Where do we want to go?):**
- Developing strategic alternatives at Corporate level (Stability, Expansion, Retrenchment, Combination), Business/SBU level (Porter's Generic Strategies: Cost Leadership, Differentiation, Focus), and Functional level.

- Evaluating alternatives using portfolio tools (BCG, GE Nine-Cell, SWOT/TOWS) and selecting the best strategy based on Suitability, Feasibility, and Acceptability.

• **3. Strategy Implementation (How do we get there?):**

- Translating chosen strategy into action via organizational structure design, resource allocation, functional policies, corporate culture alignment, and strategic leadership.

• **4. Strategic Evaluation and Control (How are we doing?):**

- Monitoring ongoing assumptions and actual performance against strategic targets (via Strategic Surveillance, Premise Control, Implementation Control, and Special Alert Control) and taking corrective actions.

### Q12. Explain the value chain analysis process.

**Answer:** Introduced by Michael E. Porter (1985), **Value Chain Analysis (VCA)** views a firm as a sequential chain of value-creating activities that transform inputs into outputs valued by customers, generating a profit margin (Total Value Created minus Total Cost of Activities).

**1. Classification of Value Chain Activities:**

• **Primary Activities (Direct production & delivery):** (i) Inbound Logistics, (ii) Operations, (iii) Outbound Logistics, (iv) Marketing & Sales, and (v) After-Sales Service.

• **Support Activities (Infrastructure & enablement):** (i) Firm Infrastructure, (ii) Human Resource Management, (iii) Technology Development (R&D;), and (iv) Procurement.

**2. Step-by-Step Value Chain Analysis Process:**

• **Step 1: Identify and Segregate Activities:** Break down the firm's operations into distinct primary and support sub-activities.

• **Step 2: Allocate Costs and Identify Cost Drivers:** Assign operating costs and assets to each activity and identify the structural cost drivers (scale, capacity utilization, linkages) to find cost-reduction opportunities.

• **Step 3: Identify Value-Creating (Differentiation) Drivers:** Determine which activities create superior buyer value (quality, speed, innovation, customer service).

• **Step 4: Evaluate Internal and External Linkages:** Analyze how performance in one activity (e.g., quality inspection in Procurement) reduces costs in another (e.g., fewer warranty claims in Service), as well as vertical linkages with suppliers and distribution channels.

• **Step 5: Benchmark Against Competitors & Optimize:** Compare the firm's activity costs and value against industry best practices and outsource non-core activities where external specialists are cheaper.

### Q13. What is meant by internal development? Describe the methods for internal development.

**Answer:**

**Meaning of Internal Development (Organic Growth):**

Internal development is a strategic method of pursuing corporate growth by building upon and expanding a firm's own internal resources, capabilities, R&D;, production capacity, and sales force—rather than acquiring or merging with external companies. It allows the firm to maintain cultural uniformity, spread investment outlays gradually over time, and avoid acquisition premiums.

**Methods for Internal Development:**

• **1. In-House Research & Development (R&D;) and New Product Development:** Investing in internal laboratories and product engineering teams to invent and commercialize new products or upgrade existing products.

• **2. Capacity Expansion (Greenfield / Brownfield Investment):** Building new manufacturing plants, opening new company-owned retail showrooms/branches, and purchasing modern machinery to increase output.

• **3. Market Penetration and Geographic Expansion:** Intensifying advertising, expanding the internal sales force, launching e-commerce channels, and entering new regional or export markets using internal marketing teams.

• **4. Internal Human Capital & Core Competency Building:** Developing proprietary training programs, knowledge management systems, and cross-functional innovation teams (intrapreneurship / internal corporate venturing) to enter new business lines organically.

### Q14. Develop a sample of strategic advantages profile (SAP) for a firm of your choice.

**Answer:** A **Strategic Advantage Profile (SAP)** is a structured summary matrix resulting from an internal appraisal that assesses a firm's key functional areas and capabilities as competitive **Strengths (+)**, **Weaknesses (-)**, or **Neutral (0)** relative to industry competitors.

Below is a sample SAP prepared for **Chaudhary Group (CG Foods — Wai Wai Noodles Division)**:

| Internal Functional Area / Capability Factor | Assessment (+ / - / 0) | Strategic Impact & Competitive Justification |
|---|---|---|
| **1. Marketing & Distribution** | **+ (Major Strength)** | Dominant brand equity of 'Wai Wai' across Nepal & South Asia with an extensive nationwide dealer and retail network. |
| **2. Operations & Manufacturing** | **+ (Strength)** | Multiple automated manufacturing plants across provinces (Bardiya, Nawalparasi, Sunsari) yielding economies of scale. |
| **3. Research & Development (R&D;)** | **0 (Neutral)** | Moderate product line extensions (cup noodles, seasoning variants), but limited breakthrough health-food innovation. |
| **4. Finance & Accounting** | **+ (Strength)** | Strong operating cash flows, high creditworthiness, and easy access to institutional bank financing. |
| **5. Supply Chain / Raw Materials** | **- (Weakness)** | Heavy reliance on imported wheat flour, palm oil, and packaging materials exposes margins to exchange rate & import shocks. |
| **6. Human Resources & Culture** | **+ (Strength)** | Experienced professional management team blended with strong entrepreneurial leadership. |

### Q15. Explain the criteria for strategic choice.

**Answer:** According to Gerry Johnson, Kevan Scholes, and Richard Whittington, evaluating and selecting the best strategic option (**Strategic Choice**) relies on three primary criteria—known as the **SFA Framework (Suitability, Feasibility, and Acceptability)**:

• **1. Suitability (Does the strategy fit our situation?):**
Assesses the strategic logic of an option—whether it addresses the key external opportunities and threats (ETOP), builds on the firm's internal strengths and core competencies, remedies its weaknesses (SAP), and aligns with the company's vision and mission.

• **2. Feasibility (Can the strategy be implemented with our resources?):**
Evaluates whether the organization possesses (or can realistically obtain) the practical resources and capabilities required to execute the strategy—including *Financial feasibility* (cash flow and capital funding), *Human/Skill feasibility*, *Technological feasibility*, and *Supply-chain/Market access*.

• **3. Acceptability (Will stakeholders support the expected return and risk?):**
Measures whether the expected performance outcomes of the strategy meet the expectations of powerful stakeholders across three dimensions:
- *Return:* Profitability (ROI, NPV, EPS, shareholder value creation).
- *Risk:* Financial ratios, liquidity, sensitivity to downturns, and break-even risk.
- *Stakeholder Reactions:* Political and cultural acceptability to shareholders, lenders, employees, regulators, and society.

### Q16. What are the basic requirements for effective implementation of strategy? Describe.

**Answer:** Strategy formulation is only half the battle; translating a chosen strategy into action requires the following basic requirements (aligned with McKinsey's 7S and implementation frameworks):

• **1. Matching Organizational Structure to Strategy:** As Alfred Chandler proved ("Structure follows Strategy"), a new strategy requires an appropriate structure (functional, divisional, SBU, matrix, or network) with clear lines of authority and accountability.

• **2. Adequate Resource Allocation (Budgeting):** Financial capital, physical facilities, technology, and human talent must be reallocated toward strategic priorities through strategy-driven capital and operating budgets.

• **3. Supportive Corporate Culture:** Shared values, norms, and beliefs of employees must be aligned with the new strategic direction (e.g., fostering an innovation culture for a differentiation strategy or a cost-conscious culture for cost leadership).

• **4. Effective Strategic Leadership:** Top leaders must communicate the vision clearly, champion change, overcome organizational resistance/inertia, and role-model commitment.

• **5. Functional Policies, Procedures, and Information Systems:** Standard operating procedures (SOPs), ERP/management information systems, and functional policies (in HR, Marketing, Operations) must guide daily employee decisions.

• **6. Performance-Linked Reward and Incentive Systems:** Tying bonuses, promotions, and recognition directly to the achievement of annual strategic milestones and Balanced Scorecard KPIs.

### Q17. Write the meaning of strategic audit and explain its importance.

**Answer:**

**Meaning of Strategic Audit:**

A strategic audit is a comprehensive, systematic, and objective diagnostic examination of an organization's entire strategic management process—covering its external environment, internal resources, mission/objectives,

corporate/business/functional strategies, implementation systems, and performance outcomes—using a standardized checklist of strategic questions.

**Importance of Strategic Audit:**

• **1. Holistic Diagnostic Tool:** Provides top management and the Board of Directors with a complete 360-degree health check of the company's strategic posture, ensuring no critical external threat or internal weakness is overlooked.
• **2. Early Detection of Strategic Drift:** Identifies gaps between formulated plans and actual execution before minor operational problems escalate into financial crises.
• **3. Objective Resource Allocation:** Evaluates whether capital and managerial talent are being invested in high-return strategic business units rather than decaying legacy operations.
• **4. Improved Corporate Governance & Accountability:** Gives independent directors, institutional investors, and auditors a structured benchmark to evaluate executive management's stewardship.
• **5. Continuous Organizational Learning:** Institutionalizes feedback from past successes and failures to refine future strategy formulation.

## Group 'C': Long Answer Questions (Attempt any THREE — ALL Solved) [3 x 10 = 30]

### Q18. 'GE matrix is useful tool for evaluating strategy in an organization than BCG'. Justify.

**Answer:**

**1. Overview of BCG and GE Nine-Cell Portfolio Matrices:**

Both the Boston Consulting Group (BCG) Growth-Share Matrix and the General Electric (GE) / McKinsey Nine-Cell Matrix are corporate portfolio analysis tools used by multi-business firms to evaluate their Strategic Business Units (SBUs) and allocate capital across them.

• The **BCG Matrix** is a **2x2 (4-cell)** matrix that plots SBUs using only two single-variable dimensions: *Market Growth Rate* (High/Low) and *Relative Market Share* (High/Low), classifying SBUs into Stars, Cash Cows, Question Marks, and Dogs.
• The **GE Nine-Cell Matrix** is a **3x3 (9-cell)** matrix developed by McKinsey & Co. for General Electric that plots SBUs along two composite, multi-factor dimensions: **Industry Attractiveness** (High, Medium, Low) and **Business Unit Competitive Strength** (Strong, Average, Weak), prescribing three broad zones:
- *Green Zone (3 top-left cells):* Invest / Grow (Build)
- *Yellow Zone (3 diagonal middle cells):* Hold / Selectively Invest
- *Red Zone (3 bottom-right cells):* Harvest / Divest.

**2. Justification: Why GE Nine-Cell Matrix is Superior to the BCG Matrix:**

• **1. Multi-Factor Composite Measurement vs. Single-Variable Oversimplification:**
- BCG assumes that Market Growth Rate is the sole indicator of industry attractiveness and Relative Market Share is the sole indicator of competitive strength.
- In reality, a fast-growing industry may have intense price wars and low profit margins, and a low-market-share niche firm can be highly profitable through differentiation.
- The **GE Matrix** overcomes this by using weighted composite scores of multiple factors: *Industry Attractiveness* includes market size, growth rate, industry profitability, Porter's five forces, and PESTEL factors; *Business Strength* includes market share, brand equity, R&D;, product quality, cost structure, and managerial competence.
• **2. Inclusion of Middle/Intermediate Categories (3x3 vs. 2x2):**
- BCG forces an artificial High/Low dichotomy with no middle ground.
- The GE Matrix introduces **Medium** Industry Attractiveness and **Average** Business Strength, providing 9 finer gradations that reflect real-world business positions much more accurately.
• **3. Richer, More Nuanced Strategic Prescriptions:**
- Instead of branding every low-share/low-growth unit a "Dog" to be liquidated, the GE Matrix provides tailored prescriptions across 9 cells (e.g., selective specialization, niche defense, managed harvesting) and visually displays both total industry size (circle area) and the SBU's market share (pie slice inside the circle).
• **4. Applicability Across Diverse Industries and Lifecycle Stages:**
- BCG fails in mature or differentiated industries where scale economies are less important; the GE Matrix allows managers to customize factor weights to fit each specific industry context.

### Q19. Explain the Porter's Five Forces Model of environmental analysis.

**Answer:**

Developed by Harvard Professor **Michael E. Porter (1979)**, the **Five Forces Model** is the most widely used framework for analyzing the competitive structure and long-term profit attractiveness of an industry (task environment). Porter argues that industry profitability is determined not just by direct competitors, but by the collective strength of five competitive forces:

• **1. Threat of New Entrants (Potential Competitors):**

New entrants bring additional production capacity, desire to gain market share, and downward pressure on prices. The height of **Barriers to Entry** determines this threat:

- Economies of scale and high capital requirements
- Brand loyalty and high customer switching costs
- Access to distribution channels and proprietary technology/patents
- Government licensing, tariffs, and regulatory barriers
- Expected retaliation by entrenched incumbents.

**• 2. Bargaining Power of Buyers (Customers):**

Powerful buyers can force down industry prices, demand higher quality or extra services, and play competitors against each other. Buyer power is high when:

- A few large buyers purchase a high volume of the seller's output
- The industry's products are standardized or undifferentiated commodities
- Switching costs to another supplier are low
- Buyers pose a credible threat of *backward vertical integration*.

**• 3. Bargaining Power of Suppliers:**

Powerful suppliers can squeeze industry profitability by raising input prices or reducing the quality/reliability of raw materials and components. Supplier power is high when:

- The supplier industry is dominated by a few companies (oligopoly)
- The supplied input is unique, differentiated, or has no close substitutes
- The focal industry is not an important customer of the supplier group
- Suppliers pose a credible threat of *forward vertical integration*.

**• 4. Threat of Substitute Products or Services:**

Substitutes are products from outside the industry that perform the same or similar function for customers (e.g., tea vs. coffee, airlines vs. high-speed rail, plastic vs. glass packaging). Substitutes place a ceiling on the prices firms in an industry can charge. The threat is high when substitutes offer an attractive price-performance trade-off and buyer switching costs are low.

**• 5. Intensity of Rivalry Among Existing Competitors:**

The central force where incumbent firms jockey for position using price cuts, advertising battles, new product introductions, and extended warranties. Rivalry is fierce when:

- Numerous or equally balanced competitors exist in the industry
- Industry growth is slow or stagnant (turning competition into a fight for market share)
- Fixed costs or storage costs are high
- Product differentiation and switching costs are low
- High **exit barriers** (specialized assets, labor agreements, emotional attachment) keep unprofitable firms trapped in the industry.

**Q20. Strategic leadership is vital in effective implementation of strategies. Comment on the statement with roles and responsibilities of strategic leadership.**

**Answer:**

**Comment on the Statement:**

The statement is completely valid. Even the most brilliantly formulated strategy will fail during execution without strong **strategic leadership**. Strategy implementation requires mobilizing thousands of employees, overcoming organizational inertia and political resistance, reallocating scarce resources, and reshaping corporate culture. Strategic leaders—the CEO, top management team, and SBU heads—act as the bridge between strategic intent (formulation) and operational reality (implementation).

**Key Roles and Responsibilities of Strategic Leadership in Strategy Implementation:**

**• 1. Determining Strategic Direction and Articulating a Compelling Vision:**

Strategic leaders craft and continuously communicate a clear long-term vision and mission so employees at every level understand why the strategy matters and how their daily work contributes to it.

**• 2. Exploiting and Maintaining Core Competencies:**

Leaders ensure that the firm's unique resources and core capabilities (R&D;, customer service, supply-chain agility) are nurtured, shared across business units, and leveraged in executing the strategy.

**• 3. Developing Human Capital and Empowering Managers:**

Recruiting, training, coaching, and retaining talented executives and frontline workers, and delegating authority to middle and

operational managers so they can make fast, aligned decisions.

• **4. Building and Sustaining an Effective Organizational Culture:**
Shaping shared values, entrepreneurial mindset, customer orientation, and adaptability through visible leadership behavior, stories, symbols, and reward systems.

• **5. Emphasizing Ethical Practices and Corporate Governance:**
Instilling integrity, transparency, regulatory compliance, and ethical codes of conduct to protect the organization's reputation and stakeholder trust.

• **6. Establishing Balanced Organizational Controls (Balanced Scorecard):**
Designing balanced financial and non-financial controls (customer, internal process, learning & growth metrics) that keep execution on track without stifling innovation and autonomy.

**Q21. Write the concept of strategic control and explain its process.**

**Answer:**

**Concept of Strategic Control:**

Strategic control is a specialized form of organizational control that focuses on monitoring and evaluating the strategic management process itself to ensure that the strategy is being implemented as planned, that the underlying environmental and internal premises remain valid, and that the results produced by the strategy are achieving corporate goals.

Unlike traditional operational (post-action) control—which asks *"Did we do things right?"* after the year ends—strategic control is forward-looking (feedforward and concurrent) and asks *"Are we still moving in the right direction?"* across four types (Schreyogg & Steinmann): **Premise Control**, **Strategic Surveillance**, **Special Alert Control**, and **Implementation Control**.

**Step-by-Step Process of Strategic Control:**

• **Step 1: Determine What to Control (Identify Key Strategic Areas):**
Select the critical success factors, strategic milestones, and key environmental assumptions (premises) that are most vital to the success of the strategy.

• **Step 2: Set Strategic Standards and Benchmarks (KPIs):**
Establish clear quantitative standards (ROI, market share, cost per unit, revenue growth) and qualitative standards (brand reputation, product quality, employee morale, innovation rate) using tools like the Balanced Scorecard.

• **Step 3: Measure Actual Performance and Monitor Environmental Premises:**
Continuously gather internal performance data through Management Information Systems (MIS/ERP) and strategic audits while scanning the external environment for changes in premises or sudden crises.

• **Step 4: Compare Actual Performance and Premises Against Standards:**
Evaluate whether actual outcomes fall within acceptable tolerance limits of the planned targets.

• **Step 5: Analyze Deviations and Diagnose Root Causes:**
If a significant variance occurs, determine whether it was caused by poor operational execution, unrealistic targets, or an unexpected shift in the external environment.

• **Step 6: Take Corrective Action:**
Reformulate implementation plans, reallocate resources, revise functional tactics, or—if environmental premises have fundamentally changed—update the strategy itself.

## Group 'D': Comprehensive Answer / Case / Situation Analysis [4 x 5 = 20]

**Q22. Case Study: XTP Nepal (Distributor of Chinese Footwear Brand ABC / Xtep) XTP Nepal was established in Kathmandu with NRs 8 million investment by Ganesh Gautam and Shiva Shrestha (previously operating GS Enterprises wholesale footwear) after winning the distributorship of Chinese brand ABC (parallel to Adidas, Nike, Anta, Li-Ning, Xtep, Skechers) in 2013. Adidas has 4 showrooms with brand warranty; Anta closed 5 showrooms and partnered with KTM brand; Li-Ning dropped 4 showrooms; Rainbow has 85 outlets across Nepal plus 1 Newroad retail shop. Local unbranded footwear copies branded designs. Language barrier (Chinese/English) and quarterly business plan/progress reporting requirements from China HQ stress Gautam and Shrestha. Kingsway showroom had frustrating sales due to limited selection, while Bouddha showroom performed above expectations; Kingsway was shifted to Kumaripati (Lalitpur). a. Conduct an industry analysis for XTP Nepal. b. Prepare generic strategy for XTP Nepal to increase the sales of Xtep in Nepal. c. What organizational structure do you suggest to fit the ABC branding in Nepal? Why? d. XTP is planning the central control of all strategies of all outlets. What components of strategic information systems do you suggest to XTP? Why?**

**Answer (Comprehensive Case Study Solution):**

**a. Industry Analysis for XTP Nepal (Porter's Five Forces & Competitive Environment Analysis):**

• **1. Intensity of Competitive Rivalry (High):** The branded sportswear and footwear industry in Nepal is fiercely competitive.

Global giants like Adidas (4 exclusive showrooms with brand warranty) and Skechers compete at the premium end, while Chinese/regional peers like Anta (partnered with local garment producer KTM CTY), Li-Ning, and Rainbow (85 multi-brand outlets nationwide) compete aggressively for urban youth and sports enthusiasts.

• **2. Threat of Substitutes and Counterfeit/Imitation Products (Very High):** Numerous local branded and unbranded footwear makers in Kathmandu rapidly copy the designs of international brands and sell them at much lower prices, eroding the sales of genuine imported brands.

• **3. Bargaining Power of Suppliers / Brand Principal (High):** ABC/Xtep Head Office in China holds strong bargaining power over XTP Nepal—demanding strict quarterly business plans, progress reports, and company-owned showroom expansion, and has already replaced distributors in other countries.

• **4. Bargaining Power of Buyers (Moderate to High):** Nepalese urban customers are brand-conscious (responding to sports trends and celebrity endorsements like cricketers Sandeep Lamichhane and Gyanendra Malla) but demand wide variety/selection and value-for-money; as seen at the Kingsway showroom, buyers walk away if product assortment is narrow.

• **5. Industry Growth & Opportunities (Favorable):** Rising sports culture, youth population, fitness awareness, and expanding commercial hubs (such as Bouddha and Kumaripati in Lalitpur) offer strong growth potential for complete sports apparel and footwear lines.

**b. Recommended Generic Strategy for XTP Nepal to Increase Sales of Xtep/ABC in Nepal:**

Based on Michael Porter's Generic Strategies, XTP Nepal should pursue an **Integrated Cost-Conscious Differentiation Strategy (Best-Cost Provider / Focused Differentiation Strategy)**:

• **1. Product-Mix Differentiation (Broad Assortment of Footwear + Sportswear):** The Kingsway showroom failed because buyers had limited selection. XTP must stock the full range of shoes alongside sports apparel (t-shirts, trousers, socks, bags, footballs, volleyballs) in all showrooms.

• **2. Brand Positioning & Warranty Assurance ("International Quality at Mid-Premium Price"):** Position ABC/Xtep between ultra-expensive Adidas/Nike and cheap local counterfeits by offering an official authenticity guarantee and warranty (matching Adidas's trust factor) so buyers prefer genuine Xtep over local design copies.

• **3. Strategic Showroom Location & Hybrid Distribution:** Focus company-owned flagship showrooms in high-footfall youth/residential hubs like Bouddha and Kumaripati, while leveraging GS Enterprises' existing wholesale dealer network across major cities outside Kathmandu (similar to Rainbow's 85-outlet reach) to achieve volume scale.

**c. Suggested Organizational Structure to Fit ABC Branding in Nepal & Justification:**

• **Recommended Structure:** A **Functional-cum-Divisional (Hybrid / Geographic-Channel) Organizational Structure** separating the Wholesale Business (GS Enterprises) from XTP Nepal's Branded Retail Operations, supported by dedicated functional departments:

- **Managing Directors (Gautam & Shrestha)** at the top.
- **Three Core Operational Divisions/Departments:**

1. *Retail Showroom Operations Division* (Store Managers for Newroad, Bouddha, Kumaripati, and future outlets).
2. *Wholesale & Dealer Distribution Division* (managing regional multi-brand outlets).
3. *Professional Corporate Support Functions:* (i) **International Liaison, Planning & Reporting Officer** (fluent in English/Chinese and skilled in preparing quarterly business plans and financial reports for China HQ), (ii) **Marketing & Brand Promotion Unit**, and (iii) **Finance & Inventory Control Unit**.

• **Why This Structure Fits Best:** Currently, Mr. Gautam is distracted by the legacy wholesale business while Mr. Shrestha is overwhelmed handling ABC issues and China HQ's quarterly reports with limited language and technical planning skills. Separating wholesale from retail and hiring a professional Planning & International Liaison Manager eliminates executive stress, satisfies China HQ's professional standards, and prevents loss of the distributorship.

**d. Recommended Components of Strategic Information Systems (SIS) for Central Control of All Outlets:**

To centrally control strategies, sales, and inventory across all showrooms and wholesale outlets, XTP Nepal should implement an integrated **Retail Enterprise Resource Planning (ERP) & Strategic Information System** comprising four key components:

• **1. Centralized Point-of-Sale (POS) & Real-Time Inventory Management System:** Connects Newroad, Bouddha, Kumaripati, and warehouse stock in real time using barcode/SKU scanning. *Why:* Immediately identifies fast-moving vs. slow-moving sizes/designs, prevents stockouts (solving the "limited selection" problem seen in Kingsway), and automates stock transfers between outlets.

• **2. Financial & Quarterly Management Reporting System (MIS / BI Dashboard):** Automatically compiles daily and quarterly sales, gross margins, outlet-wise profitability, and budget vs. actual variance reports in English. *Why:* Directly solves Mr. Gautam and Mr. Shrestha's biggest headache—generating accurate, professional quarterly business plans and progress reports required by ABC Head Office in China.

• **3. Customer Relationship Management (CRM) & Loyalty System:** Tracks customer preferences, repeat purchases, warranty claims, and promotional campaign effectiveness. *Why:* Builds brand loyalty against local counterfeit footwear.

• **4. Supply Chain & Import Procurement Portal (SCM):** Links XTP Nepal's order forecasting directly with ABC China's supply portal. *Why:* Reduces lead times for seasonal sportswear and footwear launches.

# PAPER 2: TRIBHUVAN UNIVERSITY — BNK 221: CORPORATE FINANCING DECISION

**BBA / Eighth Semester | April 2026 | Full Marks: 100 | Pass Marks: 50 | Time: 3 Hrs**

## Group 'A': Brief Answer Questions [10 x 2 = 20]

### Q1. Define Corporate financing decisions.

**Answer:** Corporate financing decisions are the strategic decisions made by financial managers concerning the raising of short-term and long-term funds from internal and external sources (debt, preferred stock, and common equity) to finance the firm's assets and operations, aiming to minimize the weighted average cost of capital (WACC) and maximize firm value.

### Q2. State any two factors to be considered while choosing an appropriate source of short-term financing.

**Answer:** Two primary factors to consider while selecting a source of short-term financing are:
• **1. Effective Annual Cost (Interest Rate & Fees):** Comparing the true after-tax effective annual interest cost across trade credit, commercial paper, and bank loans.
• **2. Availability, Flexibility & Matching Principle:** Ensuring the credit source is reliably available when seasonal working capital needs peak and matches the self-liquidating cash conversion cycle.

### Q3. Differentiate between bankruptcy liquidation and reorganization.

**Answer:**
• **Bankruptcy Liquidation (e.g., Chapter 7):** The permanent shutdown of an insolvent company where a court-appointed trustee sells off all the firm's assets and distributes the cash proceeds to creditors in strict order of legal priority (when the firm is worth more dead than alive).
• **Reorganization (e.g., Chapter 11):** A court-supervised restructuring of a financially distressed company's debt obligations (extending maturities, reducing interest rates, or swapping debt for equity) so the business continues operating as a going concern (when the firm is worth more alive than dead).

### Q4. Write the meaning of private equity fund.

**Answer:** A private equity (PE) fund is a pooled investment vehicle (typically organized as a Limited Partnership managed by a General Partner) that raises capital from institutional and accredited investors (Limited Partners) to acquire significant equity stakes in private companies or take public companies private (via Venture Capital, Growth Equity, or Leveraged Buyouts) with a 5-to-10-year investment horizon.

### Q5. Define initial conversion premium.

**Answer:** Initial conversion premium is the percentage (or Rupee amount) by which the conversion price ($P_c$) of a newly issued convertible bond or convertible preferred stock exceeds the current market price of the common stock ($P_0$) at the time of issuance:
**Conversion Premium (%) = [ ($P_c$ - $P_0$) / $P_0$ ] x 100** (typically set between 15% and 30%).

### Q6. What is warrants?

**Answer:** A warrant is a long-term company-issued call option (often attached as an "equity sweetener" to a bond or preferred stock issue) that gives the holder the right to purchase a stated number of shares of the issuing company's common stock at a specified exercise price within a stated period of time.

### Q7. Mention the major types of leases.

**Answer:** The major types of leases are:
• **1. Operating Lease** (short-term, cancellable, maintained by lessor)
• **2. Financial / Capital Lease** (long-term, non-cancellable, fully amortized)
• **3. Sale and Leaseback** (selling an owned asset and leasing it back immediately)
• **4. Leveraged Lease** (three-party lease involving lessee, lessor, and long-term lender)
• **5. Direct Lease** (acquiring use of a new asset directly from manufacturer/lessor).

### Q8. If you borrow Rs 100,000 face value at a nominal rate of 10 percent, discount interest, for 6 months, Calculate effective annual rate.

**Answer (Step-by-Step Numerical Solution):**
**Given:**
• Face value of loan (Principal) = Rs 100,000
• Nominal annual interest rate (i) = 10% = 0.10 (Discount interest)

• Loan maturity = 6 months = 0.5 year (m = 12 / 6 = 2 periods per year)

**Step 1: Calculate 6-Month Discount Interest and Usable Proceeds:**
• 6-Month Interest = Rs 100,000 x 0.10 x (6 / 12) = **Rs 5,000**
• Net Usable Funds = Rs 100,000 - Rs 5,000 = **Rs 95,000**
• 6-Month Periodic Effective Rate (r_p) = Rs 5,000 / Rs 95,000 = **0.052632 (5.2632%)**

**Step 2: Calculate Effective Annual Rate (EAR) and Nominal Annual Rate (APR):**
• **Compounded Effective Annual Rate (EAR):**
EAR = (1 + r_p)^m - 1 = (1 + 0.052632)^2 - 1 = (1.052632)^2 - 1 = 0.108033 = **10.80%**
• **Simple Annualized Percentage Rate (APR):**
APR = r_p x 2 = 5.2632% x 2 = **10.53%**
*(Note: If 10% is treated as a full-year discount rate: EAR = 0.10 / (1 - 0.10) = **11.11%**; if 10% is treated as a 6-month rate: EAR = (1 + 10/90)^2 - 1 = **23.46%**. Showing all ensures full marks!)*

***EXAM HINT TO REMEMBER:** For a 6-month discount loan at 10% p.a.: 6-month interest = 10% / 2 = 5% (Rs 5,000), usable proceeds = Rs 95,000, so 6-month rate = 5/95 = 5.2632%. Then EAR = (1 + 5/95)^2 - 1 = 10.80% (and APR = 2 x 5.2632% = 10.53%).*

**Q9. Mr. Shyam Thapa, a dissatisfied stockholder of the Lumbini Company, desires representation on the board. The Lumbini Company, which has 7 directors, has 1 million shares outstanding. How many shares would Mr. Thapa have to control to be assured of 1 directorship under a majority rule (non-cumulative) voting system?**

**Answer (Step-by-Step Numerical Solution):**
**Given:**
• Total shares outstanding (N) = 1,000,000 shares
• Total number of directors (D) = 7 directors
• Voting system: **Majority Rule (Non-Cumulative / Straight Voting) System**

**Step 1: Formula Under Majority Rule (Non-Cumulative) Voting:**
Under majority (non-cumulative) voting, each director position is voted on separately, so a shareholder who controls **more than 50% of the shares** elects **every single director** (whether 1 director or all 7 directors).
Minimum Shares Required = (N / 2) + 1

**Step 2: Substitute the Values:**
Minimum Shares Required = (1,000,000 / 2) + 1 = 500,000 + 1 = **500,001 shares**

*(Note for comparison: Had the company used Cumulative Voting, the minimum shares for 1 director would have been [(1,000,000 x 1) / (7 + 1)] + 1 = 125,001 shares.)*

***EXAM HINT TO REMEMBER:** Read the voting system carefully! Under MAJORITY (Non-Cumulative) voting, you ALWAYS need 50% + 1 shares = (N / 2) + 1 = 500,001 shares to guarantee even 1 director seat! Under CUMULATIVE voting, you need [(N x d) / (D + 1)] + 1.*

**Q10. Determine the annual percentage interest cost for 2/15 net 60 term of sale assuming that the firm does not take the cash discount but pays on the 40th day.**

**Answer (Step-by-Step Numerical Solution):**
**Given:**
• Credit terms: 2/15, net 60 -> Discount rate (d) = 2% = 0.02; Discount period (DP) = 15 days
• Actual payment day (CP_actual) = 40th day

**Step 1: Formula for Annual Percentage Cost (APC) of Not Taking Discount:**
APC = [ d / (1 - d) ] x [ Days in Year / (Actual Payment Day - Discount Period) ]

**Step 2: Calculate Using 360-Day Year (and 365-Day Year):**
• **Assuming 360 days in a year:**
APC = [ 0.02 / (1 - 0.02) ] x [ 360 / (40 - 15) ]
APC = (0.02 / 0.98) x (360 / 25) = 0.020408 x 14.4 = 0.293878 = **29.39%**
*(Compounded EAR = (1 + 0.02/0.98)^(360/25) - 1 = (1.020408)^14.4 - 1 = **33.77%**)*
• **Assuming 365 days in a year:**
APC = (0.02 / 0.98) x (365 / 25) = 0.020408 x 14.6 = **29.80%**
*(Compounded EAR = (1.020408)^14.6 - 1 = **34.31%**)*

***EXAM HINT TO REMEMBER:** Even though the stated credit period is 60 days, the firm actually paid early on the 40th day! Always use the ACTUAL payment day (40 - 15 = 25 days of costly credit) in the denominator—paying on day 40 instead of day 60 after*

*missing the discount makes the annualized percentage cost HIGHER (29.39%)!*

## Group 'B': Short Answer Questions (Attempt any SIX — ALL Solved) [6 x 5 = 30]

**Q11. What do you mean by going private? Write the motivation for going private.**

**Answer:**

**Meaning of Going Private:**

"Going private" refers to a transaction in which a publicly traded corporation's entire common stock is purchased by a small group of investors—typically existing senior management (in a Management Buyout, MBO) and/or a private equity firm (in a Leveraged Buyout, LBO)—so that the company's shares are delisted from the stock exchange and are no longer traded by the general public.

**Motivations (Reasons) for Going Private:**

• **1. Elimination of High Regulatory and Listing Compliance Costs:** Saves substantial annual outlays on securities regulator filings, public audits, quarterly reporting, shareholder meetings, and stock exchange listing fees.
• **2. Freedom from Short-Term Market Myopia:** Public managers face intense pressure to meet quarterly earnings targets. Going private allows management to focus on long-term strategic restructuring, R&D;, and capital investments without worrying about daily stock price fluctuations.
• **3. Alignment of Ownership and Management Incentives (Reduced Agency Costs):** When managers become significant equity owners, agency conflicts disappear and managers work aggressively to cut waste and boost cash flows.
• **4. Tax Shield Benefits from High Leverage ( in LBOs):** Financing the buyout with substantial debt creates large tax-deductible interest shields.
• **5. Confidentiality and Protection from Hostile Takeovers:** Private companies do not have to disclose sensitive strategic and financial data to competitors and are immune from hostile takeover bids.

**Q12. Describe the factors affecting of determining capital structure in a business organization.**

**Answer:** A firm's capital structure decision (the mix of debt and equity) is governed by several internal and external factors:
• **1. Stability and Predictability of Sales and Cash Flows:** Firms with stable, predictable revenues (e.g., utilities, consumer staples) can safely carry higher fixed-interest debt than cyclical firms.
• **2. Asset Structure (Tangibility of Assets):** Firms with large general-purpose tangible fixed assets (real estate, machinery) have strong collateral to secure low-cost long-term debt, whereas R&D;/intangible-heavy firms rely more on equity.
• **3. Operating Leverage:** A firm with high fixed operating costs (high operating leverage) already has high business risk and should use less debt (lower financial leverage) to keep total risk manageable.
• **4. Corporate Tax Rate:** Because interest on debt is tax-deductible, firms in higher income-tax brackets gain a larger debt tax shield and favor higher leverage.
• **5. Profitability and Internal Cash Generation (Pecking Order):** Highly profitable firms with abundant retained earnings often require less external debt financing.
• **6. Management Attitude and Control Considerations:** Conservative managers or founders who fear losing voting control or violating debt covenants prefer debt or internal equity cautiously depending on control dilution.
• **7. Credit Rating, Lender Attitudes, and Capital Market Conditions:** Prevailing interest rates and bank/rating agency benchmarks cap how much debt a firm can issue without a rating downgrade.

**Q13. Delphi shoes Company has warrants issued that allow the holder to purchase 4 shares of stock for a total of Rs 100 each for each warrant. Currently the market price per share of the company is Rs 20. Investor hold the following probabilistic benefit about the stock 6 month hence. [Market price per share: Rs 15 (Prob 0.30), Rs 20 (Prob 0.25), Rs 25 (Prob 0.20), Rs 30 (Prob 0.25)] a. What is the present theoretical value of warrant? b. What is the expected value of stock price 6 month hence? c. What is the expected theoretical value of warrant 6 month hence? [1+2+2]**

**Answer (Step-by-Step Numerical Solution):**

**Given:**

• Number of shares purchased per warrant (N) = 4 shares
• Total exercise price for 4 shares per warrant = Rs 100 => **Exercise price per share (E) = Rs 100 / 4 = Rs 25 per share**
*(Formula for Theoretical Value of Warrant: TV = (P_0 x N) - Total Exercise Cost = (P_0 - E) x N)*
• Current market price per share (P_0) = Rs 20

**a. Present Theoretical Value of Warrant [1 Mark]:**

• Formula TV = (P_0 - E) x N = (20 - 25) x 4 = (20 x 4) - 100 = 80 - 100 = **-Rs 20**
• Since a warrant cannot have a negative market/exercise value, its effective **Present Theoretical Value (Floor Value) = Rs 0**.

**b. Expected Value of Stock Price 6 Months Hence, E(P) [2 Marks]:**
E(P) = Sum [ P_i x Prob(P_i) ]
E(P) = (Rs 15 x 0.30) + (Rs 20 x 0.25) + (Rs 25 x 0.20) + (Rs 30 x 0.25)
E(P) = 4.50 + 5.00 + 5.00 + 7.50 = **Rs 22.00 per share**

**c. Expected Theoretical Value of Warrant 6 Months Hence [2 Marks]:**
• **Method 1 (Probability-Weighted Expected Warrant Value Across All 4 Stock Price States — Exact Finance Method):**
- If P = Rs 15 (Prob 0.30): TV = max(15 x 4 - 100, 0) = max(-40, 0) = **Rs 0**
- If P = Rs 20 (Prob 0.25): TV = max(20 x 4 - 100, 0) = max(-20, 0) = **Rs 0**
- If P = Rs 25 (Prob 0.20): TV = max(25 x 4 - 100, 0) = max(0, 0) = **Rs 0**
- If P = Rs 30 (Prob 0.25): TV = max(30 x 4 - 100, 0) = max(120 - 100, 0) = **Rs 20**
- **Expected Theoretical Value E(TV)** = (0 x 0.30) + (0 x 0.25) + (0 x 0.20) + (20 x 0.25) = **Rs 5.00**
• **Method 2 (Evaluated Directly at Expected Stock Price E(P) = Rs 22.00):**
- Formula TV at E(P) = (22 x 4) - 100 = 88 - 100 = **-Rs 12.00** (or Minimum Floor Value = **Rs 0**).

***EXAM HINT TO REMEMBER:*** *Read 'purchase 4 shares of stock for a total of Rs 100 for each warrant' carefully: Total Exercise Price = Rs 100 for 4 shares (i.e. E = Rs 25/share). In Part (c), write BOTH Method 1 (E(TV) = 20 x 0.25 = Rs 5.00) and Method 2 (Formula TV = 4(22) - 100 = -Rs 12 -> Floor Rs 0) so you get full marks under either marking scheme!*

### Q14. Explain the concept and importance of venture capital.

**Answer:**
**Concept of Venture Capital:**
Venture capital (VC) is long-term, active risk equity (or convertible security) capital provided by professional institutional venture capital funds to young, rapidly growing, innovative private companies that have high growth potential but lack the operating history and tangible collateral needed to obtain conventional bank loans or access public capital markets.

**Importance of Venture Capital:**
• **1. Seed & Early-Stage Capital for Innovation:** Enables entrepreneurs to commercialize new technologies, software, biotech, and business models.
• **2. Value-Added Managerial Mentorship:** Venture capitalists sit on the board, assist in strategic planning, recruit key executives, and connect startups with customers, suppliers, and investment banks.
• **3. Patience and Risk Absorption:** Unlike bank loans that demand monthly interest from day one, VC shares business risk and waits 5–7 years to realize capital gains via an IPO or trade sale.
• **4. Certification Effect:** Backing from a respected VC fund signals credibility and helps the startup attract follow-on financing and top talent.
• **5. Macroeconomic Growth and Employment Generation:** Drives entrepreneurship, productivity, tax revenues, and skilled job creation.

### Q15. Discuss merits and demerits of long term debt.

**Answer:**
**Merits (Advantages) of Long-Term Debt:**
• **1. Tax Deductibility of Interest:** Interest expense reduces taxable income, providing a valuable tax shield and lowering the effective after-tax cost of debt: k_dt = k_d(1 - T).
• **2. Preservation of Ownership and Voting Control:** Creditors have no voting rights in general meetings, so existing shareholders retain 100% control.
• **3. Positive Financial Leverage (Trading on Equity):** When Return on Assets (ROA) exceeds the interest rate on debt, fixed-cost debt magnifies EPS and ROE for common stockholders.
• **4. Limited Claim on Profits:** Bondholders do not share in super-normal profits when the company is highly successful.

**Demerits (Disadvantages) of Long-Term Debt:**
• **1. Increased Financial Risk and Bankruptcy Threat:** Fixed interest and principal payments must be made regardless of earnings; default can trigger foreclosure or bankruptcy liquidation.
• **2. Restrictive Indenture Covenants:** Lenders impose strict limits on dividends, further borrowing, working capital, and asset sales.
• **3. Fixed Maturity & Sinking Fund Cash Drain:** Debt must be redeemed at maturity, requiring periodic cash outflows.
• **4. Loss of Future Financing Flexibility:** Mortgaging assets today reduces emergency borrowing capacity during future downturns.

### Q16. The management of Seti Food Company Limited decided to buy a small equipment taking a loan Rs 100,000 for 3 years from Nepal Bank Limited. The loan bears an annual interest of 10 percent and calls for equal annual installment payments at the end of each of the 3 years. a. Calculate amount of annual payment. b. Prepare Loan amortization schedule. c. Calculate equal monthly installment (EMI). [1+3+1]

**Answer (Step-by-Step Numerical Solution):**

**Given:**

• Loan Principal (P_0) = Rs 100,000; Term (n) = 3 years; Annual interest rate (i) = 10% = 0.10

**a. Calculation of Equal Annual Installment Payment (PMT) [1 Mark]:**

• Present Value Interest Factor of Annuity, PVIFA(10%, 3 years) = [ 1 - (1 + 0.10)^(-3) ] / 0.10 = **2.486852**

• Annual Installment (PMT) = P_0 / PVIFA(10%, 3 yrs) = Rs 100,000 / 2.486852 = **Rs 40,211.48** (or Rs 40,211.11 using 2.4869)

**b. Loan Amortization Schedule (3 Years) [3 Marks]:**

| Year | Beginning Principal (Rs) | Annual Installment (PMT) (Rs) | Interest @ 10% (Col 2 x 0.10) (Rs) | Principal Repayment (Col 3 - Col 4) (Rs) | Ending Balance (Col 2 - Col 5) (Rs) |
|---|---|---|---|---|---|
| 1 | 100,000.00 | 40,211.48 | 10,000.00 | 30,211.48 | 69,788.52 |
| 2 | 69,788.52 | 40,211.48 | 6,978.85 | 33,232.63 | 36,555.89 |
| 3 | 36,555.89 | 40,211.48 | 3,655.59 | 36,555.89 | 0.00 |
| **Total** | — | **120,634.44** | **20,634.44** | **100,000.00** | — |

**c. Calculation of Equal Monthly Installment (EMI) [1 Mark]:**

• Number of monthly payments (n_m) = 3 x 12 = **36 months**

• Monthly interest rate (i_m) = 10% / 12 = **0.0083333 (0.8333% per month)**

• PVIFA(0.8333%, 36 months) = [ 1 - (1 + 0.10/12)^(-36) ] / (0.10/12) = **30.991236**

• **Equal Monthly Installment (EMI)** = Rs 100,000 / 30.991236 = **Rs 3,226.72 per month**

*(Note: Simple division of annual installment by 12 = Rs 40,211.48 / 12 = Rs 3,350.96.)*

***EXAM HINT TO REMEMBER:*** *In a Loan Amortization Table, Ending Balance in Year 3 MUST be 0.00, and Total Principal Repayment MUST equal the original loan (Rs 100,000). For Monthly EMI, convert n = 3 x 12 = 36 months and r = 10%/12 = 0.8333%!*

### Q17. Define Financial distress. Also write the common causes and signs of financial distress in companies. [2+3]

**Answer:**

**1. Definition of Financial Distress [2 Marks]:**

Financial distress is a severe financial condition in which a business cannot generate sufficient operating cash flows to meet its maturing financial obligations (such as interest payments, principal repayments, wages, and supplier bills) as they fall due, forcing the firm toward asset sales, debt restructuring, or legal insolvency/bankruptcy.

**2. Common Causes and Warning Signs of Financial Distress [3 Marks]:**

• **Common Causes of Financial Distress:**

1. *Excessive Financial Leverage (Over-Borrowing):* High fixed interest obligations that cannot be sustained when sales dip.
2. *Poor Working Capital & Liquidity Management:* Over-investment in slow-moving inventory, uncollectible receivables, or financing long-term fixed assets with short-term loans.
3. *High Operating Leverage & Declining Industry Demand:* Heavy fixed operating costs combined with intense competition, technological obsolescence, or macroeconomic downturns.
4. *Managerial Incompetence & Unprofitable Expansion:* Overpaying for acquisitions or poor cost control.

• **Early Warning Signs (Symptoms) of Financial Distress:**

1. Persistent negative operating cash flows and chronic working capital deficits.
2. Delaying payments to trade creditors (stretching accounts payable 30–90 days past due) and defaulting on bank interest.
3. Dividend reductions or complete omission of dividend payments.
4. Deteriorating financial ratios (falling current ratio, interest coverage ratio < 1, rising debt-to-equity ratio) and credit rating downgrades.
5. Resignation of key executives/auditors, plant shutdowns, and distress sales of core assets.

## Group 'C': Long Answer Questions (Attempt any THREE — ALL Solved) [3 x 10 = 30]

### Q18. Explain the methods of issuing securities. Also discuss the process of initial public offering (IPO).

**Answer:**

**1. Methods of Issuing Securities in the Primary Market:**

• **(a) Public Issue via Prospectus (IPO & FPO):** Selling securities directly to the general public at large through an Initial Public Offering (first-time public sale by an unlisted company) or a Follow-on Public Offering (seasoned issue by an already listed company).

• **(b) Rights Offering (Privileged Subscription):** Offering new shares to existing shareholders on a pro-rata basis at a discounted subscription price to honour their preemptive rights.
• **(c) Private Placement:** Selling an entire issue of bonds or equity directly to a small group of institutional investors (insurance companies, mutual funds, pension funds, venture capital funds) without a public prospectus—saving time and flotation costs.
• **(d) Offer for Sale / Bought-Out Deal:** Selling the securities en bloc to an investment bank or issue house, which then resells them to the public.
• **(e) Book Building & Shelf Registration:** Discovering the issue price through institutional bidding (book building) or registering a master prospectus to sell securities in tranches over time (shelf registration).

**2. Step-by-Step Process of Initial Public Offering (IPO):**
• **Step 1: Internal Corporate Authorization:** Pass a Board of Directors resolution and obtain shareholder approval at the General Meeting to convert/issue public capital.
• **Step 2: Appointment of Issue Manager and Underwriter:** Select a licensed merchant banker (Issue Manager) to conduct due diligence, structure the issue, and underwrite the offering.
• **Step 3: Credit Rating and Preparation of Draft Prospectus:** Obtain an IPO grading/credit rating from a rating agency (e.g., ICRA/CARE Nepal) and draft the prospectus containing audited financials, risk factors, and use of proceeds.
• **Step 4: Filing & Approval from Securities Regulator (SEBON / SEC):** Submit the registration statement and prospectus to the regulator; address regulatory comments during the cooling-off period until final approval is granted.
• **Step 5: Publication of Offer Document & Opening of Subscription:** Publish the approved prospectus/notice in national daily newspapers and open public subscription (e.g., through C-ASBA participating banks).
• **Step 6: Issue Closure, Basis of Allotment & Refund:** Close subscription, verify applications, allot shares (minimum 10 units per applicant / lottery system in Nepal), and unblock/refund unallotted money.
• **Step 7: Demat Credit & Secondary Market Listing:** Credit allotted shares into investors' Demat accounts at CDSC and list the shares on the Stock Exchange (NEPSE) for secondary trading.

**Q19. The Himal Company wishes to evaluate two plans, leasing and borrowing to purchase, for financing an oven. The firm is in the 40 percent tax bracket. Lease: The company can lease the oven under a 5 years lease requiring annual end of year payment of Rs 14,000. Purchase: The oven costs Rs 50,000 and will have a 5-year life. It will be depreciated under straight line depreciation method. The total purchase price will be financed by a 5-years, 10% loan requiring equal annual end of year payments. The salvage value is zero. a. Calculate the present value of leasing. b. Calculate the present value of purchasing. c. Which alternative, lease or purchase, would you recommend? Why? [4+4+2]**

**Answer (Step-by-Step Numerical Solution):**
**Given:**
• Corporate tax rate (T) = 40% = 0.40
• Annual lease payment (L) = Rs 14,000 at the end of Years 1 to 5
• Purchase cost of oven ($I_0$) = Rs 50,000; Useful life (n) = 5 years; Salvage value (SV) = Rs 0
• Before-tax cost of debt ($k_d$) = 10% => **After-tax cost of debt ($k_{dt}$) = 10% x (1 - 0.40) = 6%**
• Present Value Annuity Factor at 6% for 5 years:
PVIFA(6%, 5 yrs) = [ 1 - (1.06)^(-5) ] / 0.06 = **4.212364** (or 4.2124)

**a. Present Value of Cost of Leasing [4 Marks]:**
• Annual after-tax lease cash outflow = L x (1 - T) = Rs 14,000 x (1 - 0.40) = **Rs 8,400 per year**
• **Present Value of Leasing** = Rs 8,400 x PVIFA(6%, 5 yrs)
= Rs 8,400 x 4.212364 = **Rs 35,383.86** (or Rs 35,384.16 using 4.2124)

**b. Present Value of Cost of Purchasing (Borrowing to Buy) [4 Marks]:**
• Annual Straight-Line Depreciation = (Rs 50,000 - 0) / 5 = **Rs 10,000 per year**
• Annual Depreciation Tax Shield = Rs 10,000 x 0.40 = **Rs 4,000 per year**
• *(Note on Loan Schedule: A 5-year 10% loan of Rs 50,000 has annual installment PMT = 50,000 / PVIFA(10%, 5) = 50,000 / 3.790787 = Rs 13,189.87. Because the loan is discounted at the after-tax debt rate $k_{dt}$ = 6%, the Present Value of loan principal + after-tax interest payments is identically equal to the initial Rs 50,000 purchase price!)*
• **Present Value of Purchasing** = Initial Cost - PV of Depreciation Tax Shields
= Rs 50,000 - [ Rs 4,000 x PVIFA(6%, 5 yrs) ]
= Rs 50,000 - (Rs 4,000 x 4.212364)
= Rs 50,000 - Rs 16,849.46 = **Rs 33,150.54** (or Rs 33,150.40 using 4.2124)

| Year | Loan Payment (Rs) | Interest @ 10% (Rs) | Depreciation (Rs) | Tax Shield 40% x (Int+Dep) (Rs) | Net Cash Outflow (Col 2 - Col 5) | PV @ 6% (Rs) |
|---|---|---|---|---|---|---|
| 1 | 13,189.87 | 5,000.00 | 10,000.00 | 6,000.00 | 7,189.87 | 6,782.90 |
| 2 | 13,189.87 | 4,181.01 | 10,000.00 | 5,672.41 | 7,517.46 | 6,690.51 |
| 3 | 13,189.87 | 3,280.13 | 10,000.00 | 5,312.05 | 7,877.82 | 6,614.37 |
| 4 | 13,189.87 | 2,289.15 | 10,000.00 | 4,915.66 | 8,274.21 | 6,553.95 |
| 5 | 13,189.87 | 1,199.08 | 10,000.00 | 4,479.63 | 8,710.24 | 6,508.81 |
| **Total PV of Purchasing** | | | | | | **33,150.54** |

**c. Recommendation [2 Marks]:**
Since the Present Value of Cost of Purchasing (**Rs 33,150.54**) is lower than the Present Value of Cost of Leasing (**Rs 35,383.86**) by **Rs 2,233.32** (Net Advantage to Leasing = -Rs 2,233.32), Himal Company should **BORROW AND PURCHASE the oven**.

***EXAM HINT TO REMEMBER:*** *Whether you use the quick formula [PV(Buy) = 50,000 - 4,000 x PVIFA(6%, 5) = Rs 33,150.54] or the full 5-year loan amortization schedule table above, both give the EXACT same answer: Rs 33,150.54! Use the quick formula to verify your table in 10 seconds!*

**Q20. The Nepal textile company plans to raise an additional Rs 10 million through rights offering. Current market price of the company is Rs 250. It has 200,000 shares outstanding. Stockholders are offered a new share at a price of Rs 100 each. You are asked to answer following questions. a. How many new shares will have to be sold to raise required funds? b. How many rights will be required to purchase a new share? c. What will be theoretical value of a right and ex-right price? d. Mr. Ram Shrestha's total assets consist of 100 shares of Nepal Textile Company and cash balance of Rs 15,000. Prepare statements of Mr. Shrestha's total assets before rights offerings. e. Prepare statement showing Mr. Shrestha total assets after right offerings for each of these course of actions if he: i. Sell all his rights. ii. Exercise all his rights. [1+1+3+2+3]**

**Answer (Step-by-Step Numerical Solution):**
**Given:**
• Funds to be raised = Rs 10,000,000 (Rs 10 million)
• Rights-on market price per share ($P_0$) = Rs 250
• Existing shares outstanding ($N_0$) = 200,000 shares
• Subscription price per new share ($P_s$) = Rs 100

**a. Number of New Shares to Be Sold ($N_1$) [1 Mark]:**
$N_1$ = Funds to be Raised / Subscription Price ($P_s$) = Rs 10,000,000 / Rs 100 = **100,000 new shares**

**b. Number of Rights Required to Purchase One New Share (#) [1 Mark]:**
# = Existing Shares ($N_0$) / New Shares ($N_1$) = 200,000 / 100,000 = **2 rights per new share**

**c. Theoretical Value of a Right ($V_r$) and Ex-Rights Price ($P_e$) [3 Marks]:**
• **Theoretical Value of One Right ($V_r$):**
$V_r = (P_0 - P_s) / (\# + 1) = (250 - 100) / (2 + 1) = 150 / 3 =$ **Rs 50 per right**
• **Ex-Rights Price of Stock ($P_e$):**
$P_e = P_0 - V_r$ = Rs 250 - Rs 50 = **Rs 200 per share**

**d. Statement of Mr. Ram Shrestha's Total Assets BEFORE Rights Offering [2 Marks]:**
• Value of 100 shares @ Rs 250 (rights-on price) = 100 x Rs 250 = Rs 25,000
• Cash balance = Rs 15,000
• **Total Assets Before Rights Offering = Rs 25,000 + Rs 15,000 = Rs 40,000**

**e. Statement of Mr. Ram Shrestha's Total Assets AFTER Rights Offering [3 Marks]:**
• **(i) If He Sells All His 100 Rights (@ Rs 50 each):**
- Value of 100 existing shares at Ex-Rights Price (Rs 200) = 100 x Rs 200 = Rs 20,000
- Existing Cash + Cash from selling 100 rights = Rs 15,000 + (100 x Rs 50) = Rs 20,000
- **Total Assets After Selling Rights = Rs 20,000 + Rs 20,000 = Rs 40,000**
• **(ii) If He Exercises All His 100 Rights:**
- Number of new shares purchased with 100 rights = 100 rights / 2 rights per share = **50 new shares**
- Cash paid to subscribe to 50 new shares @ Rs 100 = 50 x Rs 100 = Rs 5,000

- Value of Total Shares (100 + 50 = 150 shares) @ Ex-Rights Price (Rs 200) = 150 x Rs 200 = Rs 30,000
- Remaining Cash Balance = Rs 15,000 - Rs 5,000 = Rs 10,000
- **Total Assets After Exercising Rights = Rs 30,000 + Rs 10,000 = Rs 40,000**

***EXAM HINT TO REMEMBER:*** *In Rights Offering wealth statements (parts d & e), Total Wealth MUST stay identical (Rs 40,000) Before, After Selling Rights, and After Exercising Rights! If your three totals match Rs 40,000, your math is 100% verified!*

**Q21. The Delta company wishes to borrow Rs 100,000 for 1 year. It must choose one of the following alternatives. a. 9 percent loan on a collect basis, with face value due at end. b. 8.4 percent loan on a discount basis, with face value due at the end. c. 6 percent loan on an add on basis, with equal quarterly payments required on the initial face value. Calculate the effective cost of each alternative, using annual compounding for the first two alternatives and quarterly compounding for the last. Which one alternative would you recommend? [3+3+3+1]**

**Answer (Step-by-Step Numerical Solution):**
**Given:** Face value of loan = Rs 100,000; Maturity = 1 year.

**a. 9% Loan on a Collect (Simple Interest) Basis [3 Marks]:**
• Interest paid at year-end = Rs 100,000 x 0.09 = Rs 9,000; Usable funds = Rs 100,000
• **Effective Annual Cost (EAR_a)** = Rs 9,000 / Rs 100,000 = **9.00%**

**b. 8.4% Loan on a Discount Basis [3 Marks]:**
• Discount interest deducted upfront = Rs 100,000 x 0.084 = Rs 8,400
• Usable proceeds received = Rs 100,000 - Rs 8,400 = Rs 91,600
• **Effective Annual Cost (EAR_b)** = Interest / Usable Funds = 8,400 / 91,600 = 0.091703 = **9.17%**

**c. 6% Add-On Loan with Equal Quarterly Payments (Quarterly Compounding) [3 Marks]:**
• Total Add-On Interest for 1 year = Rs 100,000 x 0.06 = Rs 6,000
• Total Amount Repayable over 4 quarters = Rs 100,000 + Rs 6,000 = Rs 106,000
• Equal Quarterly Installment (PMT) = Rs 106,000 / 4 = **Rs 26,500 per quarter**
• **Exact TVM / IRR Method:**
Rs 100,000 = Rs 26,500 x PVIFA(r_q, 4 quarters) => PVIFA(r_q, 4) = 100,000 / 26,500 = **3.773585**
Solving for quarterly periodic rate r_q (by interpolation between 2% [3.8077] and 3% [3.7171]):
r_q = **2.3722% per quarter** (or nominal APR = 2.3722% x 4 = **9.49%**)
**Effective Annual Rate (Quarterly Compounding, EAR_c)** = (1 + 0.023722)^4 - 1 = **9.83%**
*(Note: Using standard Approximate Add-On Formula: Nominal APR = (2 x m x I) / [P x (m + 1)] = (2 x 4 x 6,000) / [100,000 x (4 + 1)] = 48,000 / 500,000 =* ***9.60%*** *[quarterly rate = 2.40%], and Compounded EAR = (1 + 0.024)^4 - 1 =* ***9.95%****.)*

**d. Recommendation [1 Mark]:**
Comparing the effective annual costs: Alternative (a) = **9.00%**, Alternative (b) = **9.17%**, and Alternative (c) = **9.83% (or 9.95%)**. Delta Company should select **Alternative (a) (9% Collect Basis Loan)** because it has the lowest effective annual cost (**9.00%**).

***EXAM HINT TO REMEMBER:*** *An Add-On installment loan looks cheap (6% stated rate), but because you repay principal every quarter, your average usable balance is only about half the loan! Quick formula for Add-On APR = (2 x m x I) / [P x (m + 1)] = (2 x 4 x 6,000) / (100,000 x 5) = 9.60%, and quarterly compounded EAR = (1 + 0.096/4)^4 - 1 = 9.95% (or exact IRR = 9.83%).*

## Group 'D': Comprehensive Answer / Case / Situation Analysis [20 Marks]

**Q22. Case Study: The Himalaya Herbal Company (HHC) Bank loan under line of credit has risen to Rs 250,000 (carrying 8% interest), and HHC is 30–60 days late paying trade creditors. HHC decides to raise Rs 500,000 long-term capital: Alternative 1: Sell 8% convertible bonds, convertible into 100 shares per Rs 1,000 bond (conversion price = Rs 10/share). Alternative 2: Sell 8% debentures, each Rs 1,000 bond carrying 100 warrants to buy common stock at Rs 10. Current Balance Sheet: Current liabilities = Rs 400,000; Common stock (par Re 1) = Rs 100,000 (100,000 shares); Retained earnings = Rs 50,000; Total assets = Total claims = Rs 550,000. Income Statement: Sales = Rs 1,100,000; All costs except interest = Rs 990,000; EBIT = Rs 110,000; Interest = Rs 20,000; EBT = Rs 90,000; Taxes (40%) = Rs 36,000; Net income = Rs 54,000; Shares = 100,000; EPS = Rs 0.54; P/E = 15.83x; Market price = Rs 8.55. a. Discuss the reasons for using convertible bond and bond with warrants to raise long-term fund. b. Show the new balance sheet before and after conversion of the convertible bonds. c. Show the new balance sheet before and after exercise of warrant. d. What is the earnings per share of each alternative, if it is assumed that profits before interest and taxes will be 20 percent of total assets? e. Which of the three alternatives would you recommend? [4+4+4+6+2]**

**Answer (Comprehensive Case Study Solution):**

**a. Reasons for Using Convertible Bonds and Bonds with Warrants [4 Marks]:**

• **1. Lower Coupon Rate ("Equity Sweetener"):** Because HHC is small, rapidly growing, and currently stretched on short-term credit, straight long-term debt would demand a very high interest rate. Attaching conversion rights or warrants allows HHC to borrow at a modest 8% rate.

• **2. Selling Common Stock at a Premium Over Current Market Price:** HHC's stock currently trades at Rs 8.55, whereas the conversion/warrant exercise price is Rs 10.00 (a 17% premium), issuing fewer shares than an immediate common stock sale at Rs 8.55.

• **3. Replacing Urgent Short-Term Bank Debt & Trade Payables:** Raising Rs 500,000 pays off the Rs 250,000 8% bank credit line (and overdue trade creditors) while funding growth.

• **4. Future Equity Inflow (Under Warrants):** When the 50,000 warrants are exercised at Rs 10, HHC receives an additional Rs 500,000 in fresh cash while retaining the Rs 500,000 debentures.

**Working Notes for Balance Sheets (Parts b & c):**

• Number of Rs 1,000 bonds issued = Rs 500,000 / Rs 1,000 = **500 bonds**.

• New shares upon Conversion (Alt 1) or Warrant Exercise (Alt 2) = 500 bonds x 100 shares = **50,000 new shares** at Rs 10/share (Par Re 1 -> Common Stock increases by Rs 50,000; Paid-in Capital/Share Premium increases by 50,000 x Rs 9 = Rs 450,000).

• **Application of the Rs 500,000 Raised:**

- **Standard Case (Retiring Rs 250,000 Bank Loan + Rs 250,000 Asset Expansion):** Out of Rs 500,000 raised, Rs 250,000 pays off the 8% bank loan (reducing Current Liabilities from Rs 400,000 to **Rs 150,000** and eliminating the Rs 20,000 bank loan interest), and the remaining Rs 250,000 increases Total Assets from Rs 550,000 to **Rs 800,000**. Upon exercise of warrants, an additional Rs 500,000 cash enters, raising Total Assets to **Rs 1,300,000**.

- *(Alternative Textbook Assumption where Current Liabilities stay Rs 400,000 and Total Assets become Rs 550,000 + Rs 500,000 = Rs 1,050,000 [and Rs 1,550,000 after warrant exercise] is also noted below!)*

**b & c. New Balance Sheets Before and After Conversion (Alt 1) and Warrant Exercise (Alt 2) [4+4 = 8 Marks]:**

| Claims / Liabilities & Equity | Current (Rs) | Alt 1 Conv: Before Conv. (Rs) | Alt 1 Conv: After Conv. (Rs) | Alt 2 Warrant: Before Ex. (Rs) | Alt 2 Warrant: After Ex. (Rs) |
|---|---|---|---|---|---|
| Current Liabilities (Bank loan paid) | 400,000 | 150,000 | 150,000 | 150,000 | 150,000 |
| 8% Convertible Bonds | 0 | 500,000 | 0 | 0 | 0 |
| 8% Debentures (Straight Debt) | 0 | 0 | 0 | 500,000 | 500,000 |
| Common Stock (Par Re 1) | 100,000 | 100,000 | 150,000 | 100,000 | 150,000 |
| Additional Paid-in Capital (Rs 9/sh) | 0 | 0 | 450,000 | 0 | 450,000 |
| Retained Earnings | 50,000 | 50,000 | 50,000 | 50,000 | 50,000 |
| **Total Assets = Total Claims** | **550,000** | **800,000** | **800,000** | **800,000** | **1,300,000** |

**d. Computation of Earnings Per Share (EPS) Under Each Alternative (EBIT = 20% of Total Assets) [6 Marks]:**

| Income Statement Items | Current | Alt 1: Before Conversion | Alt 1: After Conversion | Alt 2: Before Exercise | Alt 2: After Exercise |
|---|---|---|---|---|---|
| Total Assets (Rs) | 550,000 | 800,000 | 800,000 | 800,000 | 1,300,000 |
| EBIT (20% of Total Assets) | 110,000 | 160,000 | 160,000 | 160,000 | 260,000 |
| Less: Interest (8% on Debt) | 20,000 | 40,000 | 0 | 40,000 | 40,000 |
| Earnings Before Taxes (EBT) | 90,000 | 120,000 | 160,000 | 120,000 | 220,000 |
| Less: Taxes @ 40% | 36,000 | 48,000 | 64,000 | 48,000 | 88,000 |
| **Net Income (EAT) (Rs)** | **54,000** | **72,000** | **96,000** | **72,000** | **132,000** |
| Common Shares Outstanding | 100,000 | 100,000 | 150,000 | 100,000 | 150,000 |
| **Earnings Per Share (EPS)** | **Rs 0.54** | **Rs 0.72** | **Rs 0.64** | **Rs 0.72** | **Rs 0.88** |

*(Note: If Current Liabilities of Rs 400,000 are kept unchanged so Total Assets = Rs 1,050,000 before exercise [EBIT = Rs 210,000, Interest = 20,000 + 40,000 = Rs 60,000] and Rs 1,550,000 after warrant exercise [EBIT = Rs 310,000, Interest = Rs 60,000], then EPS Before Conv/Ex = Rs 0.90, EPS After Conv = Rs 0.76, and EPS After Warrant Exercise = Rs 1.00.)*

**e. Recommendation [2 Marks]:**

• If HHC expects continued rapid growth and needs the additional Rs 500,000 capital inflow when warrants are exercised

(which generates the **highest EPS of Rs 0.88**), it should choose **Alternative 2 (Debentures with Warrants)**.
• Conversely, if HHC wants to eliminate long-term debt and fixed interest from its balance sheet upon conversion (achieving a debt-free balance sheet with EPS of **Rs 0.64**, up from Rs 0.54 currently), it should choose **Alternative 1 (Convertible Bonds)**.

***EXAM HINT TO REMEMBER:*** *Notice how Current Interest of Rs 20,000 in the Income Statement equals 8% of the Rs 250,000 Bank Loan! Paying off that Rs 250,000 bank loan from the Rs 500,000 bond issue reduces Current Liabilities from Rs 400,000 to Rs 150,000 and increases Total Assets by the remaining Rs 250,000 (from Rs 550,000 to Rs 800,000).*

# PAPER 3: TRIBHUVAN UNIVERSITY — BNK 202: FINANCIAL DERIVATIVES

**BBA / Eighth Semester | April 2026 | Full Marks: 100 | Pass Marks: 50 | Time: 3 Hrs**

## Group 'A': Brief Answer Questions [10 x 2 = 20]

### Q1. State the types of financial derivatives markets.

**Answer:** Financial derivatives markets are classified into two main types:
• **1. Exchange-Traded Derivatives Market:** Standardized contracts (such as futures and listed options) traded on organized exchanges (e.g., CME, CBOE) with a central clearinghouse, daily marking-to-market, and zero counterparty default risk.
• **2. Over-the-Counter (OTC) Derivatives Market:** Customized, privately negotiated contracts (such as forwards, swaps, and exotic options) arranged directly between financial institutions and corporate clients without an exchange floor.

### Q2. Define the term call option.

**Answer:** A call option is a financial derivative contract that gives the buyer (holder) the **right, but not the obligation**, to **buy** a specified quantity of an underlying asset at a predetermined price (the exercise or strike price, X) on or before a specified expiration date, in exchange for paying an upfront fee called the call premium (C).

### Q3. State the meaning of minimum value of put option.

**Answer:** The minimum value (or lower bound) of a put option is the lowest theoretical price at which the put can trade without creating a riskless arbitrage opportunity:
• **For a European Put:** P_min = max[ X * e^(-rT) - S_0 , 0 ] (or max[ X / (1 + r)^T - S_0 , 0 ])
• **For an American Put:** P_min = max[ X - S_0 , 0 ] (its intrinsic value, since it can be exercised immediately).

### Q4. Write the meaning of overpriced put with one example.

**Answer:** An **overpriced put option** is a put option whose actual market price (P_mkt) exceeds its fair theoretical value (as determined by Black-Scholes, Binomial model, or Put-Call Parity) or exceeds its upper bound (X).
• **Example:** If the Black-Scholes fair value of a put option on ABC stock is Rs 18, but the put is currently trading in the market at Rs 25, the put option is overpriced by Rs 7, and an arbitrageur/trader should **sell (write) the overpriced put** and hedge with the underlying stock.

### Q5. List out any four assumptions of Black Scholes option pricing model.

**Answer:** Four core assumptions of the Black-Scholes-Merton (BSM) option pricing model are:
• **1. European-Style Exercise & No Dividends:** The option can only be exercised at expiration, and the underlying stock pays no dividends during the option's life.
• **2. Lognormal Stock Price & Constant Volatility:** Stock prices follow a continuous geometric Brownian motion with constant return volatility (sigma).
• **3. Constant Risk-Free Interest Rate:** The continuously compounded risk-free rate (r) is known and constant across all maturities.
• **4. Frictionless Markets:** There are no transaction costs or taxes, short selling is permitted without penalty, and trading is continuous.

### Q6. What is butterfly spread?

**Answer:** A **butterfly spread** is a neutral four-option combination strategy using three strike prices (X_1 < X_2 < X_3, equidistant) with the same expiration date—constructed by **buying 1 low-strike option (X_1), selling (writing) 2 middle-strike at-the-money options (X_2), and buying 1 high-strike option (X_3)**. A long butterfly spread profits when the underlying stock price stays stable near the middle strike price (X_2) and has strictly limited downside risk.

### Q7. Outline any two differences between futures and options.

**Answer:**
• **1. Obligation vs. Right (Symmetry):** In a **futures contract**, both the buyer (long) and seller (short) have a legally binding **obligation** to perform at maturity (symmetric payoff). In an **option contract**, the buyer has the **right (no obligation)** to exercise, while the seller has the obligation if assigned (asymmetric payoff).
• **2. Upfront Cost & Risk Profile:** Entering a futures contract requires no upfront premium (only a refundable margin deposit) and carries unlimited gain/loss potential for both sides; buying an option requires paying a non-refundable upfront **premium**, which caps the buyer's maximum loss at the premium paid.

### Q8. Mention two common types of cash flows exchanged in swaps.

**Answer:** Two common types of cash flows exchanged in swap contracts are:
• **1. Fixed vs. Floating Interest Cash Flows (in a Plain Vanilla Interest Rate Swap):** One party pays a fixed interest rate

while the other pays a floating benchmark rate (such as SOFR) on the same notional principal in the same currency.
• **2. Principal and Interest Cash Flows in Two Different Currencies (in a Currency Swap):** Or Equity Index Return vs. Floating Interest Cash Flows (in an Equity Swap).

### Q9. State the meaning of basis risk in hedging.

**Answer:** In futures hedging, **Basis** is defined as the difference between the Spot Price of the hedged asset and the Futures Price of the contract used: **Basis = Spot Price (S_t) - Futures Price (F_t)**.
**Basis risk** is the uncertainty (risk) that the basis will change unpredictably between the time a hedge is initiated and the time it is closed out—arising from mismatches in asset quality/grade, delivery location, or maturity date—preventing a hedge from being 100% perfect.

### Q10. You take a long futures contract on the S&P; 500 when the futures price is 1,107.40, and close it out three days later at a futures price of 1,131.75. One futures contract is for 250 the index. Ignoring interest, what are your losses/gains?

**Answer (Step-by-Step Numerical Solution):**
**Given:**
• Position: 1 Long S&P; 500 Futures Contract
• Initial entry futures price (F_0) = 1,107.40
• Closing futures price after 3 days (F_t) = 1,131.75
• Contract multiplier = $250 (or Rs 250) per index point

**Calculation of Net Gain:**
Gain = (F_t - F_0) x Index Multiplier x Number of Contracts
Gain = (1,131.75 - 1,107.40) x 250 x 1
Gain = 24.35 x 250 = **6,087.50 (Gain of $6,087.50 or Rs 6,087.50)**

***EXAM HINT TO REMEMBER:*** *Long Futures Gain/Loss = (Ending Futures Price - Beginning Futures Price) x Contract Multiplier = (1,131.75 - 1,107.40) x 250 = +6,087.50.*

## Group 'B': Short Answer Questions (Attempt any SIX — ALL Solved) [6 x 5 = 30]

### Q11. What is financial derivatives? Write its key characteristics.

**Answer:**
**Meaning of Financial Derivatives:**
A financial derivative is a bilateral financial contract or instrument whose value is **derived from** (depends upon) the price or value of an underlying asset, reference rate, or index—such as stocks, bonds, foreign currencies, commodities, market indices, or interest rates. The main types of derivatives are Forwards, Futures, Options, and Swaps.

**Key Characteristics of Financial Derivatives:**
• **1. Value Derived from an Underlying Asset:** Has no independent intrinsic value of its own; its price fluctuates directly with changes in the spot price of the underlying asset.
• **2. Future Settlement Date:** Performance or cash settlement takes place at a specified future date (or period) agreed upon today.
• **3. Zero or Small Initial Net Investment (High Leverage):** Requires little or no upfront cash outlay compared to buying the underlying asset outright, magnifying percentage returns and risks.
• **4. Zero-Sum Game Nature:** Every long position is matched by an equal and opposite short position, so one counterparty's gain equals the other's loss.
• **5. Settlement Flexibility:** Most financial derivatives can be closed out prior to maturity via an offsetting trade or settled in cash rather than physical delivery.

### Q12. Define a forward contract. Also write its essential features.

**Answer:**
**Definition of a Forward Contract:**
A forward contract is a customized, privately negotiated Over-the-Counter (OTC) agreement between two parties in which one party (the buyer, holding the **long position**) agrees to buy and the other party (the seller, holding the **short position**) agrees to sell a specific quantity and quality of an underlying asset at a fixed price (the **forward price**) on a specified future date (the **maturity/delivery date**).

**Essential Features of a Forward Contract:**
• **1. Bilateral Over-the-Counter (OTC) Negotiation:** Privately negotiated directly between two counterparties (e.g., a bank and an importer) without an exchange floor.
• **2. Complete Customization:** The contract size, underlying asset quality, delivery date, and settlement procedure are

tailored to the exact needs of the two parties.
• **3. No Upfront Premium or Daily Marking-to-Market:** At initiation, the value of a forward contract is zero and no money changes hands; gains or losses are settled only once at maturity.
• **4. Counterparty Default (Credit) Risk:** Because there is no central clearinghouse or daily margining, each party bears the credit risk that the other party may default at maturity.
• **5. Low Secondary Liquidity:** Cannot be easily transferred or closed out with a third party without the consent of the original counterparty.

**Q13. The current price of ABC stock is Rs 300 per share. The call option is available on this stock with exercise price of Rs 300 per share. The expiration date is 3 month. The call premium is Rs 10 per share. The possible ending prices are Rs 260, Rs 270, Rs 280, Rs 290, Rs 300, Rs 310, Rs 320, Rs 330, and Rs 340. a. Calculate the breakeven point of call option. b. Calculate the value of call at each ending prices. c. Calculate the gain or loss for call buyer at each ending prices.**

**Answer (Step-by-Step Numerical Solution):**
**Given:**
• Current stock price (S_0) = Rs 300; Exercise price (X) = Rs 300; Call premium (C) = Rs 10

**a. Breakeven Point (BEP) of the Call Option:**
Breakeven Stock Price (S_T*) = Exercise Price (X) + Call Premium (C)
S_T* = Rs 300 + Rs 10 = **Rs 310 per share**

**b & c. Value (Payoff) of Call at Expiration [V_C = max(S_T - 300, 0)] and Net Gain/Loss for Call Buyer [Profit = V_C - 10]:**

| Ending Stock Price (S_T) | Exercise Decision | (b) Value of Call at Expiration V_C = max(S_T - 300, 0) | Call Premium Paid (C) | (c) Net Gain / Loss for Call Buyer (V_C - 10) |
|---|---|---|---|---|
| Rs 260 | Do Not Exercise | Rs 0 | Rs 10 | -Rs 10 (Loss) |
| Rs 270 | Do Not Exercise | Rs 0 | Rs 10 | -Rs 10 (Loss) |
| Rs 280 | Do Not Exercise | Rs 0 | Rs 10 | -Rs 10 (Loss) |
| Rs 290 | Do Not Exercise | Rs 0 | Rs 10 | -Rs 10 (Loss) |
| Rs 300 (ATM) | Indifferent / No | Rs 0 | Rs 10 | -Rs 10 (Loss) |
| **Rs 310 (BEP)** | **Exercise** | **Rs 10** | **Rs 10** | **Rs 0 (Breakeven)** |
| Rs 320 | Exercise | Rs 20 | Rs 10 | **+Rs 10 (Gain)** |
| Rs 330 | Exercise | Rs 30 | Rs 10 | **+Rs 20 (Gain)** |
| Rs 340 | Exercise | Rs 40 | Rs 10 | **+Rs 30 (Gain)** |

***EXAM HINT TO REMEMBER:*** *'Value of Call at Ending Price' (part b) is the Intrinsic Payoff max(S_T - X, 0) (which is never negative!), whereas 'Gain or Loss for Call Buyer' (part c) subtracts the Rs 10 premium: Payoff - C.*

**Q14. Suppose, the current price of the Himal stock is Rs 200, the exercise price of call is Rs 180, the annual risk-free interest rate is 10% and the time to expiration is one year. a. Calculate the lower bound of European call b. Calculate arbitrage profit if call is selling at Rs 15 in the market.**

**Answer (Step-by-Step Numerical Solution):**
**Given:**
• Current stock price (S_0) = Rs 200; Exercise price (X) = Rs 180
• Annual risk-free interest rate (r) = 10% = 0.10; Time to expiration (T) = 1 year
• Market price of call option (C_mkt) = Rs 15

**a. Lower Bound (Minimum Value) of the European Call Option:**
• **Method 1 (Continuous Compounding — Standard BSM/Hull Formula):**
C_min = max[ S_0 - X * e^(-rT) , 0 ]
C_min = 200 - 180 * e^(-0.10 * 1) = 200 - (180 * 0.904837) = 200 - 162.87 = **Rs 37.13**
• **Method 2 (Annual Discrete Compounding — Standard Chance Formula):**
C_min = max[ S_0 - X / (1 + r)^T , 0 ]
C_min = 200 - (180 / 1.10) = 200 - 163.64 = **Rs 36.36**

**b. Arbitrage Profit and Strategy When Call is Selling at Rs 15 in the Market:**
• Since Market Price (Rs 15) < Lower Bound (Rs 37.13 or Rs 36.36), the call option is severely **underpriced**!
• **Arbitrage Profit Today (at t = 0):**

- Under Continuous Compounding: Arbitrage Profit = Rs 37.13 - Rs 15.00 = **Rs 22.13 per share today** (or Rs 22.13 * e^(0.10) = **Rs 24.46 at maturity**).
- Under Annual Compounding: Arbitrage Profit = Rs 36.36 - Rs 15.00 = **Rs 21.36 per share today** (or Rs 21.36 * 1.10 = **Rs 23.50 at maturity**).

• **How to Lock In the Arbitrage (Today at t = 0):**

1. **Buy 1 underpriced Call Option** for -Rs 15.00.
2. **Short-sell 1 share of Himal Stock** for +Rs 200.00.
3. **Invest the Present Value of X** (Rs 162.87 continuous or Rs 163.64 discrete) at 10% risk-free rate (which grows to exactly Rs 180 in 1 year to exercise the call and cover the short stock), pocketing the riskless net cash inflow of **Rs 22.13** (or **Rs 21.36**) immediately today!

***EXAM HINT TO REMEMBER:*** *Show BOTH continuous discounting [200 - 180*e^(-0.10) = Rs 37.13 -> Arbitrage Profit Rs 22.13] and annual discounting [200 - 180/1.10 = Rs 36.36 -> Arbitrage Profit Rs 21.36] so your answer matches any marking key!*

**Q15. Suppose that a stock is priced at Rs 80 and has a standard deviation of 30 percent. You buy a call option with an exercise price of Rs 80 that expires in 3 months. The risk-free rate is 5 percent. Determine the theoretical value of the call. Suppose that the actual call is selling for Rs 5. Suggest your investment strategy.**

**Answer (Step-by-Step Numerical Solution):**

**Given:**

• Current stock price (S_0) = Rs 80; Exercise price (X) = Rs 80
• Standard deviation (sigma) = 30% = 0.30; Time to expiration (T) = 3 / 12 = **0.25 year**
• Risk-free rate (r) = 5% = 0.05; Actual market price of call (C_mkt) = Rs 5.00

**Step 1: Compute d_1 and d_2 under Black-Scholes-Merton (BSM) Model:**

• ln(S_0 / X) = ln(80 / 80) = ln(1) = **0**
• sigma * sqrt(T) = 0.30 * sqrt(0.25) = 0.30 * 0.50 = **0.15**
• **d_1** = [ ln(S_0 / X) + (r + 0.5 * sigma^2) * T ] / (sigma * sqrt(T))
d_1 = [ 0 + (0.05 + 0.5 * 0.09) * 0.25 ] / 0.15 = (0.095 * 0.25) / 0.15 = 0.02375 / 0.15 = **0.1583 (or 0.16)**
• **d_2** = d_1 - sigma * sqrt(T) = 0.1583 - 0.15 = **0.0083 (or 0.01)**

**Step 2: Cumulative Normal Probabilities N(d_1) and N(d_2):**

• N(d_1) = N(0.1583) = **0.5629** *(Using rounded Z = 0.16: N(0.16) = 0.5636)*
• N(d_2) = N(0.0083) = **0.5033** *(Using rounded Z = 0.01: N(0.01) = 0.5040)*

**Step 3: Theoretical Value of the Call Option (C_0):**

C_0 = S_0 * N(d_1) - X * e^(-rT) * N(d_2)
• S_0 * N(d_1) = 80 * 0.5629 = **Rs 45.032**
• X * e^(-rT) * N(d_2) = 80 * e^(-0.05 * 0.25) * 0.5033 = 80 * 0.987578 * 0.5033 = 79.006 * 0.5033 = **Rs 39.764**
• **C_0** = 45.032 - 39.764 = **Rs 5.27** *(Using 2-decimal Z-table: 80(0.5636) - 79.006(0.5040) = 45.088 - 39.819 = **Rs 5.27**)*

**Step 4: Recommended Investment Strategy (When Actual Call Price = Rs 5.00):**

Since the actual market price of the call (**Rs 5.00**) is less than its theoretical fair value (**Rs 5.27**), the call option is **undervalued (underpriced) by Rs 0.27**. Therefore, the investor should **BUY the call option** (and, if hedging to lock in riskless arbitrage, short N(d_1) = 0.5629 shares of the stock per call bought).

***EXAM HINT TO REMEMBER:*** *When S_0 = X (At-The-Money), ln(S_0/X) = ln(1) = 0! That simplifies d_1 to [(r + 0.5*sigma^2)*T] / (sigma*sqrt(T)). Compare Market Call (Rs 5.00) with Fair BSM Call (Rs 5.27): Market < Fair => Underpriced => BUY Call!*

**Q16. An investor enters into a short futures position in 10 contracts in gold at a futures price of $276.50 per ounce. The size of one futures contract is 100 ounce. The initial margin per contract is $1,500, and the maintenance margin is $1,100. Suppose the futures settlement price on the first day is $278.00 per ounce, second day is $281.00 per ounce, third day is $276.00, fourth day is $280.00, and fifth day is $275.00. a. Calculate daily gain or loss, cumulative gain or loss, margin account balance, and margin call. b. If investor does not deposit margin call, what will happen?**

**Answer (Step-by-Step Numerical Solution):**

**Given:**

• Position: **Short 10 Gold Futures Contracts**
• Contract size = 100 ounces per contract => **Total Quantity = 10 x 100 = 1,000 ounces**
• Initial Futures Price (F_0) = $276.50 per ounce
• **Total Initial Margin (IM)** = 10 contracts x $1,500 = **$15,000**
• **Total Maintenance Margin (MM)** = 10 contracts x $1,100 = **$11,000**
• Since the investor holds a **SHORT position**:

- When futures price **rises**, the investor suffers a **Loss**: Daily Gain/Loss = (Previous F - Current F) x 1,000 oz.
- Whenever Ending Balance falls **below $11,000 (MM)**, a Margin Call (Variation Margin) is triggered to restore the balance to **$15,000 (IM)**.

**a. Daily Marking-to-Market Schedule (for 10 Short Contracts = 1,000 Ounces):**

| Day | Settlement Price ($/oz) | Daily Price Change ($/oz) | Daily Gain / Loss for Short ($) | Cumulative Gain / Loss ($) | Margin Balance Before Call ($) | Margin Call (Variation Margin) ($) | Ending Margin Balance ($) |
|---|---|---|---|---|---|---|---|
| Day 0 (Init) | $276.50 | — | — | $0 | $15,000 | $0 | $15,000 |
| Day 1 | $278.00 | +$1.50 | -$1,500 | -$1,500 | $13,500 | $0 (>=11,000) | $13,500 |
| **Day 2** | **$281.00** | +$3.00 | **-$3,000** | -$4,500 | **$10,500 (<11,000)** | **+$4,500** | **$15,000** |
| Day 3 | $276.00 | -$5.00 | +$5,000 | +$500 | $20,000 | $0 | $20,000 |
| Day 4 | $280.00 | +$4.00 | -$4,000 | -$3,500 | $16,000 | $0 | $16,000 |
| Day 5 | $275.00 | -$5.00 | +$5,000 | +$1,500 | $21,000 | $0 | $21,000 |

**b. What Happens If the Investor Does Not Deposit the Margin Call Money on Day 2:**
If the investor fails to deposit the $4,500 variation margin call on Day 2, the broker/clearinghouse will **immediately close out (liquidate) the investor's short position** by buying 10 gold futures contracts in the market at the prevailing price, deduct the accumulated loss ($4,500) and brokerage fees from the margin account, and return the remaining $10,500 balance to the investor.

***EXAM HINT TO REMEMBER:*** *Two critical rules for Q16: (1) Because it is a SHORT futures position, price UP = LOSS and price DOWN = GAIN! (2) On Day 2, balance drops to $10,500 (which is below Maintenance Margin $11,000), triggering a Margin Call of $15,000 - $10,500 = $4,500 to bring the account back up to Initial Margin ($15,000)!*

**Q17. Consider a long forward contract to buy a non-dividend-paying stock with a maturity of three months. The current stock price is Rs 400, and the annual risk-free interest rate for three months is 5%. a. Determine the forward price of the stock. b. If the forward price is Rs 430, calculate the arbitrage profit.**

**Answer (Step-by-Step Numerical Solution):**
**Given:**
• Current stock price (S_0) = Rs 400 (non-dividend-paying stock)
• Time to maturity (T) = 3 months = 3 / 12 = **0.25 year**
• Annual risk-free interest rate (r) = 5% = 0.05

**a. Theoretical Forward Price of the Stock (F_0):**
• **Method 1 (Continuous Compounding):**
F_0 = S_0 * e^(r * T) = 400 * e^(0.05 * 0.25) = 400 * e^(0.0125) = 400 * 1.012578 = **Rs 405.03**
• **Method 2 (Simple Periodic Interest):**
F_0 = S_0 * [ 1 + r * (3 / 12) ] = 400 * (1 + 0.0125) = **Rs 405.00**
• **Method 3 (Annual Discrete Compounding):**
F_0 = S_0 * (1 + r)^T = 400 * (1.05)^0.25 = 400 * 1.012272 = **Rs 404.91**

**b. Arbitrage Profit If Market Forward Price (F_mkt) = Rs 430:**
• Since Market Forward Price (Rs 430) > Theoretical Forward Price (Rs 405.03), the forward contract is **overpriced**.
• **Cash-and-Carry Arbitrage Strategy:**
1. **At t = 0 (Today):** Borrow Rs 400 at the 5% risk-free rate for 3 months, buy 1 share of the stock at S_0 = Rs 400, and enter a **short forward contract** to sell the stock in 3 months at Rs 430.
2. **At T = 3 Months:** Deliver the stock under the forward contract for **Rs 430.00** and repay the loan plus interest of **Rs 405.03** (or Rs 405.00 / Rs 404.91).
3. **Riskless Arbitrage Profit at Maturity:**
- Under Continuous Compounding: Rs 430.00 - Rs 405.03 = **Rs 24.97 per share**
- Under Simple Interest: Rs 430.00 - Rs 405.00 = **Rs 25.00 per share**
- Under Annual Compounding: Rs 430.00 - Rs 404.91 = **Rs 25.09 per share**

***EXAM HINT TO REMEMBER:*** *Whenever Market Forward Price (Rs 430) > Fair Forward Price (Rs 405.03), use Cash-and-Carry Arbitrage: Borrow cash -> Buy Spot Asset -> Sell (Short) Forward Contract. Arbitrage Profit = F_mkt - F_0 = 430 - 405.03 = Rs 24.97!*

## Group 'C': Long Answer Questions (Attempt any THREE — ALL Solved) [3 x 10 = 30]

**Q18. Discuss how options can be used to hedge equity risk, including the strategies involved and their advantages and limitations.**

**Answer:**

**1. Concept of Hedging Equity Risk with Options:**

Equity investors and portfolio managers face downside market risk (the risk that stock prices or the overall stock index will decline). Options provide asymmetric risk protection—functioning like an insurance policy that establishes a guaranteed floor value against market crashes while preserving upside participation when stock prices rise.

**2. Major Option Strategies Used to Hedge Equity Risk:**

• **Strategy 1: Protective Put (Portfolio Insurance: Long Stock + Long Put):**

- *Mechanism:* An investor who holds a stock (or equity portfolio) purchases a put option with strike price X.
- *Payoff at Expiration:* Value = max(S_T, X) - P_0. If the stock price crashes below X, the investor exercises the put and sells the stock at X (limiting maximum loss to S_0 - X + P_0). If the stock price surges above X, the put expires worthless and the investor enjoys unlimited upside gain minus the put premium P_0.

• **Strategy 2: Covered Call (Income & Partial Downside Cushion: Long Stock + Short Call):**

- *Mechanism:* An investor holding a stock writes (sells) an out-of-the-money call option against the stock and collects the call premium C_0.
- *Payoff:* The call premium C_0 provides a partial buffer against small declines in the stock price (reducing the effective cost basis to S_0 - C_0), in exchange for capping upside gains at the strike price X.

• **Strategy 3: Equity Collar (Zero-Cost Collar: Long Stock + Long OTM Put + Short OTM Call):**

- *Mechanism:* The investor buys a protective put at a lower strike (X_1) and finances the put premium by simultaneously writing a covered call at a higher strike (X_2 > X_1).
- *Payoff:* Locks the portfolio value within a safe band between X_1 and X_2 at virtually zero net upfront option cost.

• **Strategy 4: Hedge for a Short Stock Position (Short Stock + Long Call):**

- Buying a call option protects a short-seller against unlimited losses if the stock price rises sharply above X.

**3. Advantages of Using Options to Hedge Equity Risk:**

• **Asymmetric Protection (Downside Floor with Upside Potential):** Unlike futures/forwards (which lock in a fixed price and eliminate upside gains), a protective put insures against losses while retaining upside capital appreciation.

• **Flexibility of Strike Prices (Deductible Choice):** Investors can choose deep out-of-the-money puts (cheaper premium, higher deductible) or at-the-money puts (full protection, higher premium).

• **No Margin Calls on Long Puts:** Once the put premium is paid, the hedger faces no daily margin calls.

**4. Limitations of Hedging with Options:**

• **Upfront Premium Cost (Insurance Drag):** Buying protective puts requires paying an upfront premium that reduces portfolio return ("upside capture" drag) if the market remains calm or rises.

• **Time Decay (Theta) and Expiration Roll Risk:** Options have finite maturities and lose time value rapidly as expiration approaches, requiring costly periodic rollovers.

• **Capped Upside in Covered Calls and Collars:** Reducing hedge cost by writing calls sacrifices large bull-market gains above the call strike.

**Q19. A currency swap with semiannual payments for two years between hypothetical U.S. firm and Global Swaps, Inc. was entered into on 1 Jan 2007. Both counterparties agreed to pay fixed rate. The notional principals were EUR 10,000,000 and $9.804 million which were equivalent with exchange rate of swap initiation date. Assuming 30-day month and 360-day year. [Init Term Structure: 180d: USD 5.5%, EUR 3.8%; 360d: USD 5.5%, EUR 4.2%; 540d: USD 6.2%, EUR 4.4%; 720d: USD 6.4%, EUR 4.5%] a. Determine the fixed rates in dollars and euro. b. Assume it is 90 days into the life of swap. The exchange rate now is $0.9790/EUR and the term structures are as given below: [New Term Structure at 90d: 90d: USD 5.7%, EUR 3.9%; 270d: USD 6.1%, EUR 4.3%; 450d: USD 6.4%, EUR 4.5%; 630d: USD 6.6%, EUR 4.6%] Determine the value of the swap.**

**Answer (Step-by-Step Numerical Solution):**

**Given:**

• 2-year Semiannual Fixed-for-Fixed Currency Swap (4 payments at 180, 360, 540, and 720 days)

• Notional Principals: NP_EUR = EUR 10,000,000; NP_USD = $9,804,000 (Initial spot rate S_0 = $0.9804/EUR)

**Part (a): Determination of Fixed Swap Rates in US Dollars and Euros at Initiation (t = 0):**

• **Step 1: Compute Discount Factors B_i = 1 / [1 + R_i * (Days_i / 360)] at Initiation:**

- **US Dollar Discount Factors (B_USD):**

B_1(180d) = 1 / [1 + 0.055 * (180/360)] = 1 / 1.0275 = **0.973236**

B_2(360d) = 1 / [1 + 0.055 * (360/360)] = 1 / 1.0550 = **0.947867**

B_3(540d) = 1 / [1 + 0.062 * (540/360)] = 1 / 1.0930 = **0.914913**
B_4(720d) = 1 / [1 + 0.064 * (720/360)] = 1 / 1.1280 = **0.886525**
Sum (Sum B_USD) = 0.973236 + 0.947867 + 0.914913 + 0.886525 = **3.722541**

- **Euro Discount Factors (B_EUR):**
B_1(180d) = 1 / [1 + 0.038 * (180/360)] = 1 / 1.0190 = **0.981354**
B_2(360d) = 1 / [1 + 0.042 * (360/360)] = 1 / 1.0420 = **0.959693**
B_3(540d) = 1 / [1 + 0.044 * (540/360)] = 1 / 1.0660 = **0.938086**
B_4(720d) = 1 / [1 + 0.045 * (720/360)] = 1 / 1.0900 = **0.917431**
Sum (Sum B_EUR) = 0.981354 + 0.959693 + 0.938086 + 0.917431 = **3.796564**

• **Step 2: Calculate Semiannual and Annualized Fixed Rates:**

- **US Dollar Fixed Rate:**

Semiannual Rate (c_USD) = (1 - B_4,USD) / Sum B_USD = (1 - 0.886525) / 3.722541 = 0.113475 / 3.722541 = **0.030483 (3.0483% per half-year)**

**Annualized USD Fixed Rate (R_USD)** = 3.0483% x (360 / 180) = **6.0967% (or 6.10% p.a.)**

- **Euro Fixed Rate:**

Semiannual Rate (c_EUR) = (1 - B_4,EUR) / Sum B_EUR = (1 - 0.917431) / 3.796564 = 0.082569 / 3.796564 = **0.021748 (2.1748% per half-year)**

**Annualized EUR Fixed Rate (R_EUR)** = 2.1748% x (360 / 180) = **4.3497% (or 4.35% p.a.)**

**Part (b): Valuation of the Currency Swap 90 Days Later (Remaining Days: 90, 270, 450, 630; Spot = $0.9790/EUR):**

• **Step 1: Compute New Discount Factors at Day 90:**

- **New USD Discount Factors (B'_USD):**
B'_1(90d) = 1 / [1 + 0.057 * (90/360)] = **0.985950**
B'_2(270d) = 1 / [1 + 0.061 * (270/360)] = **0.956252**
B'_3(450d) = 1 / [1 + 0.064 * (450/360)] = **0.925926**
B'_4(630d) = 1 / [1 + 0.066 * (630/360)] = **0.896459**
Sum (Sum B'_USD) = **3.764587**
- **New EUR Discount Factors (B'_EUR):**
B'_1(90d) = 1 / [1 + 0.039 * (90/360)] = **0.990344**
B'_2(270d) = 1 / [1 + 0.043 * (270/360)] = **0.968758**
B'_3(450d) = 1 / [1 + 0.045 * (450/360)] = **0.946746**
B'_4(630d) = 1 / [1 + 0.046 * (630/360)] = **0.925497**
Sum (Sum B'_EUR) = **3.831345**

• **Step 2: Present Value of USD Fixed-Rate Leg and EUR Fixed-Rate Leg at Day 90:**

- **Value of USD Fixed Leg (per $1 Notional):**

V_USD(per $1) = c_USD * Sum B'_USD + B'_4,USD = (0.030483 * 3.764587) + 0.896459 = 0.114756 + 0.896459 = **$1.011215**

**Total Value of USD Leg (for $9,804,000 Notional)** = $9,804,000 x 1.011215 = **$9,913,952 (or $9,913,960)**

- **Value of EUR Fixed Leg (per EUR 1 Notional):**

V_EUR(per EUR 1) = c_EUR * Sum B'_EUR + B'_4,EUR = (0.021748 * 3.831345) + 0.925497 = 0.083324 + 0.925497 = **EUR 1.008821**

**Total Value of EUR Leg in Euros (for EUR 10,000,000 Notional)** = EUR 10,000,000 x 1.008821 = **EUR 10,088,210 (or EUR 10,088,227)**

**Total Value of EUR Leg Converted into USD (@ $0.9790/EUR)** = EUR 10,088,227 x 0.9790 = **$9,876,374**

• **Step 3: Net Value of the Currency Swap at Day 90:**

- **For the Party Paying EUR Fixed and Receiving USD Fixed (U.S. Firm):**

V_swap = V_USD - V_EUR(in USD) = $9,913,960 - $9,876,374 = **+$37,586 (or +EUR 38,392)**

- **For the Party Paying USD Fixed and Receiving EUR Fixed (Global Swaps, Inc.):**

V_swap = $9,876,374 - $9,913,960 = **-$37,586 (or -EUR 38,392)**

***EXAM HINT TO REMEMBER:*** *In a Fixed-for-Fixed Currency Swap, value each currency leg just like a fixed-coupon bond: V = [ Semiannual Coupon Rate x Sum(New Discount Factors) + Last Discount Factor ] x Notional Principal. Then convert the foreign leg into domestic currency using the NEW spot exchange rate ($0.9790/EUR) and subtract!*

**Q20. The current price of Himal Company stock is Rs 500 per share. Both call and put options are available on this stock with an exercise price of Rs 500. The call option premium is Rs 20 per share, and the put option**

**premium is Rs 10 per share. Both options have a maturity of one month. An investor decides to buy one call and one put on Himal Company stock because the stock price has been hovering around Rs 500 for a long time, and the investor expects a significant price movement in either direction. a. What is the name of the option position the investor is taking? b. What is the total cost of the investor's combined position? c. Calculate the profit or loss for the investor at expiration if the stock price ends at Rs 450, Rs 460, Rs 470, Rs 480, Rs 490, Rs 500, Rs 510, Rs 520, Rs 530, or Rs 540. Calculate maximum gain and loss of the strategy.**

**Answer (Step-by-Step Numerical Solution):**

**Given:**

• Current stock price (S_0) = Rs 500; Exercise price (X) = Rs 500

• Call premium (C) = Rs 20; Put premium (P) = Rs 10; Maturity = 1 month

**a. Name of the Option Position:**

Buying 1 Call Option and 1 Put Option simultaneously on the same stock with the same exercise price (Rs 500) and same expiration date is called a **Long Straddle (or Bottom Straddle)**.

**b. Total Cost of the Investor's Combined Position:**

Total Cost = Call Premium (C) + Put Premium (P) = Rs 20 + Rs 10 = **Rs 30 per share (per straddle)**.

**c. Profit or Loss at Expiration for Each Stock Price, Maximum Gain, and Maximum Loss:**

• **Maximum Possible Loss: Rs 30 per share** (the combined premium paid), which occurs if the stock price at expiration equals the strike price (S_T = Rs 500).

• **Maximum Possible Gain: Unlimited on the upside** (as S_T -> Infinity, profit -> Infinity); on the downside (if S_T falls to Rs 0), maximum gain is Rs 500 - Rs 30 = **Rs 470 per share**.

• **Breakeven Points:** Lower BEP = 500 - 30 = **Rs 470**; Upper BEP = 500 + 30 = **Rs 530**.

| Ending Stock Price (S_T) | Call Payoff max(S_T - 500, 0) | Put Payoff max(500 - S_T, 0) | Total Payoff (Rs) | Combined Cost C + P (Rs) | Net Profit / Loss (Payoff - 30) |
|---|---|---|---|---|---|
| Rs 450 | Rs 0 | Rs 50 | Rs 50 | Rs 30 | **+Rs 20 (Profit)** |
| Rs 460 | Rs 0 | Rs 40 | Rs 40 | Rs 30 | **+Rs 10 (Profit)** |
| **Rs 470 (Lower BEP)** | Rs 0 | Rs 30 | Rs 30 | Rs 30 | **Rs 0 (Breakeven)** |
| Rs 480 | Rs 0 | Rs 20 | Rs 20 | Rs 30 | -Rs 10 (Loss) |
| Rs 490 | Rs 0 | Rs 10 | Rs 10 | Rs 30 | -Rs 20 (Loss) |
| **Rs 500 (Max Loss)** | Rs 0 | Rs 0 | Rs 0 | Rs 30 | **-Rs 30 (Max Loss)** |
| Rs 510 | Rs 10 | Rs 0 | Rs 10 | Rs 30 | -Rs 20 (Loss) |
| Rs 520 | Rs 20 | Rs 0 | Rs 20 | Rs 30 | -Rs 10 (Loss) |
| **Rs 530 (Upper BEP)** | Rs 30 | Rs 0 | Rs 30 | Rs 30 | **Rs 0 (Breakeven)** |
| Rs 540 | Rs 40 | Rs 0 | Rs 40 | Rs 30 | **+Rs 10 (Profit)** |

***EXAM HINT TO REMEMBER:*** *Long Straddle = Buy 1 Call + Buy 1 Put at the SAME strike price (X = 500). Total cost = C + P = 30. Two Breakevens = X - 30 (470) and X + 30 (530). Max Loss = -30 at S_T = 500.*

**Q21. The current market price of XYZ stock is Rs 490 per share. A call option on this stock has an exercise price of Rs 500 and expires in one year. The time to expiration is divided into two binomial periods. In each period, the stock price may either rise by 20% or fall by 20%. The risk-free interest rate is 10% per annum, or 5% per period. a. Using a two-period binomial model, compute the current value of the American call option. b. If the market price of the call option is Rs 50, what investment strategy would you recommend?**

**Answer (Step-by-Step Numerical Solution):**

**Given:**

• Current stock price (S_0) = Rs 490; Exercise price (X) = Rs 500

• Number of binomial periods (n) = 2 periods (each period = 6 months)

• Up factor (u) = 1 + 0.20 = **1.20**; Down factor (d) = 1 - 0.20 = **0.80**

• Risk-free rate per period (r_p) = 5% = **0.05**

**Part (a): Current Value of the Two-Period American Call Option:**

**• Step 1: Risk-Neutral Probability (p):**

p = (1 + r_p - d) / (u - d) = (1.05 - 0.80) / (1.20 - 0.80) = 0.25 / 0.40 = **0.625**

1 - p = 1 - 0.625 = **0.375**

**• Step 2: Two-Period Binomial Stock Price Tree:**

- **Period 0 (Today):** S_0 = **Rs 490.00**
- **Period 1 (After 1st Period):**
• S_u = 490 x 1.20 = **Rs 588.00**
• S_d = 490 x 0.80 = **Rs 392.00**
- **Period 2 (At Expiration):**
• S_uu = 588 x 1.20 = **Rs 705.60** => Call Payoff C_uu = max(705.60 - 500, 0) = **Rs 205.60**
• S_ud = 588 x 0.80 = **Rs 470.40** => Call Payoff C_ud = max(470.40 - 500, 0) = **Rs 0.00**
• S_dd = 392 x 0.80 = **Rs 313.60** => Call Payoff C_dd = max(313.60 - 500, 0) = **Rs 0.00**
• **Step 3: Backward Induction to Period 1 (Checking Early Exercise for American Call):**
- **At Node U (S_u = Rs 588):**
• Value if held (unexercised): C_u(hold) = [ p * C_uu + (1 - p) * C_ud ] / (1 + r_p)
= [ (0.625 x 205.60) + (0.375 x 0) ] / 1.05 = 128.50 / 1.05 = **Rs 122.38**
• Value if exercised early at Period 1: C_u(exercise) = S_u - X = 588 - 500 = **Rs 88.00**
• Since Rs 122.38 > Rs 88.00, it is NOT optimal to exercise early! Thus, **C_u = Rs 122.38**.
- **At Node D (S_d = Rs 392):**
• C_d(hold) = [ 0.625(0) + 0.375(0) ] / 1.05 = Rs 0; C_d(exercise) = max(392 - 500, 0) = Rs 0 => **C_d = Rs 0.00**.
• **Step 4: Backward Induction to Period 0 (Today):**
C_0 = [ p * C_u + (1 - p) * C_d ] / (1 + r_p)
C_0 = [ (0.625 x 122.381) + (0.375 x 0) ] / 1.05 = 76.488 / 1.05 = **Rs 72.85**

**Part (b): Recommended Investment Strategy If Market Price of Call = Rs 50:**
Since the market price of the call option (**Rs 50.00**) is significantly lower than its fair theoretical value (**Rs 72.85**), the call option is **underpriced by Rs 22.85 per share**.
• **Recommendation:** The investor should **BUY the underpriced call option at Rs 50** (and to lock in riskless arbitrage, short-sell the delta hedge ratio h_0 = (C_u - C_d)/(S_u - S_d) = (122.38 - 0)/(588 - 392) = 122.38 / 196 = **0.6244 shares of XYZ stock** per call purchased).

***EXAM HINT TO REMEMBER:*** *Key theorem to remember in Exam: An American Call Option on a NON-dividend-paying stock is NEVER optimally exercised early (here C_u(hold) = Rs 122.38 > Early Exercise 588 - 500 = Rs 88), so American Call Value = European Call Value = Rs 72.85!*

## Group 'D': Comprehensive Answer / Case / Situation Analysis [20 Marks]

**Q22. Case Study: Naya Bank Limited (NBL) Stock & Options Today's closing price of NBL stock is S_0 = Rs 400. Call options on NBL with Exercise Price X_C = Rs 400 and 2 months to expiration sell for C = Rs 20. Put options on NBL with Exercise Price X_P = Rs 390 and 2 months to expiration sell for P = Rs 10. a. Suggest your investment strategy with stock position only in the expectation of positive economic condition. Calculate the gain on your strategy if your expectation is true and price will increases to Rs 500 per share. b. Suggest your investment strategy with option only in the expectation of positive economic condition. Calculate the gain or loss if stock price after two months increased to Rs 500 per share? Calculate the breakeven point of the investment strategy. c. Suggest your hedging investment strategy of stock with put option. Calculate the profit (or loss) from your strategy at expiration. What stock price at expiration will make you breakeven? Possible stock prices at the expiration are Rs 350, Rs 360, Rs 370, Rs 380, Rs 390, Rs 400, Rs 410, Rs 420, Rs 430, Rs 440, and Rs 450. d. Suggest your option combination strategy if you think there will be big move in stock market but you don't know the direction of movement. Calculate the profit or loss from your strategy at expiration. Possible stock prices at the expiration are Rs 350, Rs 360, Rs 370, Rs 380, Rs 390, Rs 400, Rs 410, Rs 420, Rs 430, Rs 440, and Rs 450.**

**Answer (Comprehensive Case Study Solution):**
**Given Summary:**
• NBL Current Stock Price (S_0) = Rs 400
• 2-Month Call Option: Strike Price (X_C) = Rs 400; Call Premium (C) = Rs 20
• 2-Month Put Option: Strike Price (X_P) = Rs 390; Put Premium (P) = Rs 10

**Part (a): Stock-Only Strategy Under Positive Economic Expectation:**
• **Suggested Strategy:** Take a **Long Stock Position** (buy shares of NBL in the spot market at S_0 = Rs 400 per share) to benefit from capital appreciation.
• **Gain if Stock Price Increases to S_T = Rs 500 after 2 Months:**

- **Gain per share** = S_T - S_0 = Rs 500 - Rs 400 = **+Rs 100 per share**
- **Holding Period Return (2 months)** = (Rs 100 / Rs 400) x 100 = **25%** (or annualized = 25% x 12/2 = **150% p.a.**).

**Part (b): Option-Only Strategy Under Positive Economic Expectation:**

• **Suggested Strategy:** Take a **Long Call Option Position** (buy 2-month NBL Call Options with strike X_C = Rs 400 at a premium of C = Rs 20 per share).

*(Note: For every Rs 400 equivalent to 1 share of stock, you can buy 400 / 20 = 20 call options!)*

• **Gain if Stock Price Increases to S_T = Rs 500 after 2 Months:**

- Call Payoff at S_T = Rs 500: max(500 - 400, 0) = Rs 100 per call
- **Net Gain per Call Option** = Payoff - Call Premium = Rs 100 - Rs 20 = **+Rs 80 per call option**

*(On a Rs 400 investment buying 20 calls: Total Net Gain = 20 x Rs 80 = **+Rs 1,600**!)*

- **2-Month Holding Period Return** = (Rs 80 / Rs 20) x 100 = **400%** (or annualized = 400% x 6 = **2,400% p.a.**).

• **Breakeven Point (BEP) of the Long Call Strategy:**

BEP = X_C + C = Rs 400 + Rs 20 = **Rs 420 per share**.

**Part (c): Hedging Strategy of Stock with Put Option (Protective Put Strategy):**

• **Suggested Strategy: Protective Put (Portfolio Insurance)**—Buy 1 share of NBL stock at S_0 = Rs 400 and simultaneously buy 1 Put Option (X_P = Rs 390) at P = Rs 10.

• **Total Initial Outlay:** S_0 + P = Rs 400 + Rs 10 = **Rs 410 per share**.

• **Minimum Guaranteed Floor Value at Expiration:** X_P = Rs 390 (so maximum possible loss is capped at 390 - 410 = **-Rs 20 per share**).

• **Breakeven Stock Price at Expiration (BEP):**

Profit = S_T + max(390 - S_T, 0) - 410 = 0 => **BEP = S_0 + P = Rs 400 + Rs 10 = Rs 410 per share**.

**Part (d): Option Combination Strategy for a Big Move in Either Direction (Long Strangle / Volatility Combination):**

• **Suggested Strategy:** Buy 1 Call Option (X_C = Rs 400, C = Rs 20) and Buy 1 Put Option (X_P = Rs 390, P = Rs 10).

*(Note: Because the Call strike [Rs 400] and Put strike [Rs 390] differ by Rs 10, this combination is technically a **Long Strangle**—or a Long Straddle/Strangle volatility combination!)*

• **Total Combined Premium Cost:** C + P = Rs 20 + Rs 10 = **Rs 30 per unit**.

• **Maximum Loss: -Rs 30** (for any ending stock price between Rs 390 and Rs 400).

• **Two Breakeven Points:**

- Lower Breakeven = X_P - (C + P) = Rs 390 - Rs 30 = **Rs 360 per share**
- Upper Breakeven = X_C + (C + P) = Rs 400 + Rs 30 = **Rs 430 per share**

**Combined Payoff & Profit Schedule for Part (c) Protective Put and Part (d) Long Strangle/Combination (S_T = Rs 350 to Rs 450):**

| Ending Stock Price (S_T) | Part (c): Put Payoff max(390-S_T,0) | Part (c): Protective Put Value (S_T + Put) | Part (c): Protective Put Net Profit/Loss (Value - 410) | Part (d): Call Payoff max(S_T-400,0) | Part (d): Combined Option Payoff (Call + Put) | Part (d): Option Combo Net Profit/Loss (Payoff - 30) |
|---|---|---|---|---|---|---|
| Rs 350 | Rs 40 | Rs 390 | -Rs 20 (Max Loss) | Rs 0 | Rs 40 | **+Rs 10 (Profit)** |
| Rs 360 | Rs 30 | Rs 390 | -Rs 20 (Max Loss) | Rs 0 | Rs 30 | **Rs 0 (Lower BEP)** |
| Rs 370 | Rs 20 | Rs 390 | -Rs 20 (Max Loss) | Rs 0 | Rs 20 | -Rs 10 (Loss) |
| Rs 380 | Rs 10 | Rs 390 | -Rs 20 (Max Loss) | Rs 0 | Rs 10 | -Rs 20 (Loss) |
| Rs 390 | Rs 0 | Rs 390 | -Rs 20 (Max Loss) | Rs 0 | Rs 0 | **-Rs 30 (Max Loss)** |
| Rs 400 | Rs 0 | Rs 400 | -Rs 10 (Loss) | Rs 0 | Rs 0 | **-Rs 30 (Max Loss)** |
| Rs 410 | Rs 0 | Rs 410 | **Rs 0 (BEP)** | Rs 10 | Rs 10 | -Rs 20 (Loss) |
| Rs 420 | Rs 0 | Rs 420 | **+Rs 10 (Profit)** | Rs 20 | Rs 20 | -Rs 10 (Loss) |
| Rs 430 | Rs 0 | Rs 430 | **+Rs 20 (Profit)** | Rs 30 | Rs 30 | **Rs 0 (Upper BEP)** |
| Rs 440 | Rs 0 | Rs 440 | **+Rs 30 (Profit)** | Rs 40 | Rs 40 | **+Rs 10 (Profit)** |
| Rs 450 | Rs 0 | Rs 450 | **+Rs 40 (Profit)** | Rs 50 | Rs 50 | **+Rs 20 (Profit)** |

***EXAM HINT TO REMEMBER:*** *Look closely at the strike prices in Q22: Call Strike X_C = Rs 400 (C = 20), whereas Put Strike X_P = Rs 390 (P = 10)! Because X_P = 390 != X_C = 400, (1) in Protective Put (Part c), max loss is S_0 - X_P + P = 400 - 390 + 10 = Rs 20 (for all S_T <= 390) and BEP = 400 + 10 = Rs 410; (2) in Part d (Long Strangle/Combination), Lower BEP = X_P - 30 = 390 - 30 = Rs 360 and Upper BEP = X_C + 30 = 400 + 30 = Rs 430!*

# PAPER 4: TRIBHUVAN UNIVERSITY — BNK 209: INSURANCE AND RISK MANAGEMENT

**BBA / Eighth Semester | April 2026 | Full Marks: 100 | Pass Marks: 50 | Time: 3 Hrs**

## Group 'A': Brief Answer Questions [10 x 2 = 20]

### Q1. Define the term insurance.

**Answer:** Insurance is an economic device and legal contract whereby an individual or organization (the insured) transfers the financial burden of pure, fortuitous risks to an insurance company (the insurer) in exchange for a fixed consideration called a premium, through which the losses of the unfortunate few are pooled and shared among the many exposed to the same peril.

### Q2. What do you mean by enterprise risk?

**Answer:** Enterprise risk is a comprehensive concept that encompasses all major risks faced by an organization across its entire operations—combining **Pure Risks** (property, liability, personnel), **Speculative Financial Risks** (market, credit, interest rate, foreign exchange, liquidity), **Operational Risks** (supply chain, IT/cyber failure, fraud), and **Strategic Risks** (reputation, competition, regulatory shifts)—managed holistically under Enterprise Risk Management (ERM).

### Q3. State any two reasons for regulating insurance company by NIA.

**Answer:** Two primary reasons why the **Nepal Insurance Authority (NIA / Beema Pradhikaran)** regulates insurance companies are:

- **1. Protection of Policyholders & Maintaining Insurer Solvency:** Ensuring insurers maintain adequate paid-up capital, solvency margins, and technical reserves so they can pay future claims.
- **2. Ensuring Fair Rates and Prompt Claim Settlement:** Preventing excessive or discriminatory premium pricing and protecting consumers against unfair claim denials or delays.

### Q4. Define loss forecasting and list out the common techniques of loss forecasting.

**Answer:** Loss forecasting is the analytical process used by risk managers to estimate the expected frequency (number of occurrences) and severity (Rupee size) of future losses over an upcoming planning period.

**Common Techniques of Loss Forecasting:**

- **1. Probability Analysis** (using historical frequency and severity probability distributions)
- **2. Regression Analysis** (correlating losses with physical drivers like payroll, revenue, or fleet mileage)
- **3. Forecasting Based on Loss Distributions** (e.g., Poisson and Lognormal distributions).

### Q5. Write the meaning of risk pooling.

**Answer:** Risk pooling (loss sharing) is the spreading of unexpected losses incurred by a few individuals over a large group of homogeneous exposure units so that, in the process, **average loss is substituted for actual loss**. By applying the Law of Large Numbers, risk pooling reduces objective risk for the insurer.

### Q6. State the meaning of third party insurance.

**Answer:** Third-party insurance is a liability insurance policy under which the insurer (second party) agrees to indemnify the insured (first party) against legal liability to pay compensation for bodily injury, death, or property damage caused to an innocent outsider (the **third party**)—most commonly mandatory Third-Party Motor Liability Insurance.

### Q7. Define 'endorsements' in relation to an insurance policy.

**Answer:** An **endorsement** (also called a **rider** in life insurance) is a written amendment, addendum, or legal attachment added to a standard insurance policy contract that modifies its original terms—by expanding coverage, adding extra benefits (e.g., critical illness or earthquake coverage), restricting perils, or updating the sum insured, address, or beneficiary.

### Q8. Distinguish between 'term life insurance' and 'cash value life insurance'.

**Answer:**

- **Term Life Insurance:** Provides pure death protection for a temporary specified period (e.g., 10, 15, or 20 years); pays the sum assured only if the insured dies within the term, builds **no cash surrender value (no savings element)**, and has the lowest initial premium.
- **Cash Value Life Insurance (Whole Life / Endowment):** Combines life insurance protection with a tax-advantaged **savings/investment component** that accumulates a guaranteed **cash surrender value** (which the policyholder can borrow against or receive upon surrender/maturity) and charges a higher level premium.

### Q9. Write the meaning of professional liability insurance.

**Answer:** Professional liability insurance (also known as **Errors and Omissions [E&O] Insurance** or **Professional Indemnity Insurance**, and Medical Malpractice Insurance for doctors) is a specialized liability policy that protects professionals (such as IT consultants, accountants, engineers, architects, lawyers, and physicians) against financial claims and legal defense costs arising from alleged negligence, errors, mistakes, or failure to perform professional duties.

### Q10. What is meant by personal accident insurance?

**Answer:** Personal Accident (PA) Insurance is a benefit policy that provides fixed financial compensation to the insured (or their nominee) in the event of accidental bodily injury resulting in **Accidental Death**, **Permanent Total Disability (PTD)**, **Permanent Partial Disability (PPD)**, or **Temporary Total Disability (TTD / weekly wage loss)**, as well as accidental medical expenses caused by violent, external, and visible means.

## Group 'B': Short Answer Questions (Attempt any SIX — ALL Solved) [6 x 5 = 30]

### Q11. Explain the burden of risk on society.

**Answer:** Even if an actual loss never occurs, the mere presence of pure risk imposes heavy social and economic burdens on individuals, businesses, and society as a whole:

• **1. Requirement of Larger Emergency Reserves (Idle Capital):** Without insurance, households and businesses must hold large liquid cash reserves to prepare for potential fires, medical emergencies, or lawsuits. This ties up scarce capital in low-return liquid accounts instead of productive investments, reducing economic growth.

• **2. Deprivation of Goods and Services (Deterrence of Innovation):** High-risk but socially vital goods and services (such as pharmaceuticals, aviation, power plants, or surgical procedures) may not be produced or offered if businesses fear catastrophic liability or property losses.

• **3. Mental Fear, Worry, and Anxiety:** The constant psychological stress and insecurity regarding premature death, illness, old-age poverty, or loss of one's home reduces human welfare and workplace productivity.

• **4. Direct and Indirect Cost of Uncompensated Losses:** When uninsured disasters strike, businesses go bankrupt, workers lose jobs, families fall into poverty, and the state must bear higher public welfare burdens.

### Q12. Define adverse selection. Describe the tools used by insurance companies to control the chance of adverse selection.

**Answer:**

**Definition of Adverse Selection:**

Adverse selection is the tendency of applicants with a higher-than-average chance of loss to seek and purchase insurance coverage (or buy larger coverage amounts) at standard (average) premium rates to a greater extent than average- or low-risk individuals. It stems from asymmetric information between the applicant and the insurer, and if uncontrolled, drives average claims above collected premiums.

**Tools Used by Insurance Companies to Control Adverse Selection:**

• **1. Rigorous Underwriting and Risk Classification:** Screening applicants through medical exams, health declarations, financial statements, driving records, and property inspections, and placing them into appropriate rate classes (preferred, standard, substandard, or uninsurable).

• **2. Waiting Periods and Pre-Existing Condition Exclusions:** Including a waiting period (e.g., 30 days to 24 months in health/critical illness policies) and suicide exclusion clauses so people cannot buy policies immediately before a known claim.

• **3. Deductibles, Coinsurance, and Policy Limits:** Requiring the insured to share a portion of every loss discourages high-risk applicants seeking full indemnification.

• **4. Group Underwriting Rules:** Requiring minimum participation percentages (e.g., 75%–100% of eligible employees) in employer group plans so unhealthy individuals cannot self-select alone.

• **5. Strict Enforcement of Utmost Good Faith (Voiding for Concealment/Fraud):** Cancelling policies and denying claims when applicants intentionally conceal material health or risk facts.

### Q13. Describe the role of insurance agent in insurance industry.

**Answer:** An insurance agent is a licensed intermediary authorized by an insurance company (under the Insurance Act 2079 and NIA regulations in Nepal) to solicit, negotiate, and service insurance contracts on behalf of the insurer. Their major roles are:

• **1. Public Awareness, Education, and Risk Advisory:** Educating individuals and families about financial risks (premature death, health crises, property perils) and recommending suitable insurance policies based on client needs and budget.

• **2. Field Underwriting ("Eyes and Ears of the Insurer"):** As the first point of contact who meets the applicant in person, the agent verifies the applicant's health, occupation, moral hazard, and insurable interest, preventing adverse selection and fraud at the source.

• **3. Policy Solicitation, Proposal Completion, and Premium Collection:** Assisting clients in filling out proposal forms

accurately, explaining policy terms and riders, and ensuring timely initial and renewal premium payments to prevent policy lapse.
• **4. Relationship Management and Policy Servicing:** Helping policyholders with nominations, address updates, policy loans, endorsements, and revival of lapsed policies.
• **5. Assistance in Smooth Claim Settlement:** Guiding the insured or bereaved family members through claim documentation, notification, and coordination with surveyors and the insurer's claim department during their time of distress.

**Q14. Mr. Ram Kumar Subedi, a 32 years Chartered Accountant works as an Assistant Manager in a bank. He is living with his spouse and a daughter aged 4 years. He is the single breadwinner for his family. Mr. Subedi has taken home loan from a bank for Rs 4,000,000 and a vehicle loan for Rs 2,500,000. So, he is worried that if he dies suddenly, his family will be overburdened with the loans. Being a Chartered Accountant, Mr. Subedi has good idea of investment opportunities that yields him decent returns. Suggest the most appropriate type of life insurance policy to meet his need. Justify your answer.**

**Answer (Case/Situation Analysis):**
**1. Suggested Life Insurance Policy:**
The most appropriate life insurance policy for Mr. Ram Kumar Subedi is a **Term Life Insurance Policy** (specifically, a **Level Term Life Policy** of at least Rs 6,500,000 to Rs 10,000,000 covering his working/child-dependency years, or a **Decreasing Term / Mortgage Redemption Policy** matched to his Rs 6,500,000 total bank loans), follows the classic financial planning strategy: **"Buy Term and Invest the Difference."**

**2. Justification for Recommending Term Life Insurance:**
• **(a) High Immediate Debt & Dependency Protection at Lowest Cost:** Mr. Subedi is only 32, is the sole breadwinner for his wife and 4-year-old daughter, and carries heavy liabilities totaling **Rs 6,500,000** (Rs 4,000,000 home loan + Rs 2,500,000 vehicle loan). Term life insurance provides the largest pure death benefit coverage for a very small annual premium, ensuring that if he dies prematurely, his family can immediately pay off the Rs 6.5 million bank loans and fund his daughter's upbringing.
• **(b) Superior Investment Expertise as a Chartered Accountant ("Buy Term and Invest the Difference"):** Traditional Endowment or Whole Life policies bundle insurance with low-yielding savings (typically returning only 5%–7% p.a.) and charge very high premiums. Because Mr. Subedi is a skilled Chartered Accountant with deep knowledge of high-return investment opportunities (stocks, mutual funds, bonds), he does not need an insurance company to manage his savings. By buying low-cost Term Life Insurance for pure risk protection and investing the substantial premium savings himself, he will build far greater wealth for his family.
• **(c) Recommended Riders:** He should attach an **Accidental Death & Permanent Total Disability (AD&D/PTD) Rider**, a **Critical Illness Rider**, and a **Waiver of Premium Rider** so the loan obligations are also covered if disability or severe illness stops him from working.

**Q15. Explain the role of marine insurance in international trade.**

**Answer:** Marine insurance (covering Marine Hull, Marine Cargo, and Freight against perils of the sea, inland transit, air cargo, piracy, jettison, and war/strikes under Institute Cargo Clauses A, B, and C) is the lifeblood of international trade:
• **1. Essential Requirement for Export-Import Financing (Letters of Credit):** Commercial banks will not open or negotiate an international Letter of Credit (LC) or discount export bills without a valid Marine Cargo Insurance Policy/Certificate assigned as collateral.
• **2. Protection Against Catastrophic Transit Perils:** International shipments travel thousands of miles across oceans, ports, and mountain highways (especially crucial for landlocked countries like Nepal importing via Kolkata/Visakhapatnam ports) and face perils of sinking, collision, storm, pilferage, and customs warehouse fire.
• **3. Facilitating Standardized Incoterms (CIF / CIP Trade):** Enables exporters and importers in different countries to clearly allocate transit risk under international commercial terms (such as CIF—Cost, Insurance, and Freight).
• **4. Coverage of General Average Losses:** Under maritime law, if cargo is intentionally jettisoned (thrown overboard) to save a ship in a storm, all cargo owners must share the loss proportionately (General Average); marine insurance pays these complex maritime contributions.
• **5. Expansion of Global Commerce:** By transferring catastrophic transit risk to insurers, exporters and importers can trade high-value shipments globally with confidence.

**Q16. Highlight the major issues of auto insurance.**

**Answer:** Auto (motor) insurance—covering Third-Party Liability and Comprehensive Own-Damage—faces several major operational and economic challenges (particularly in developing markets like Nepal):
• **1. High Claim Frequency and Rising Loss Ratios:** Rapid growth of vehicles, congested roads, poor road infrastructure in hilly/mountainous terrain, and reckless driving result in high accident frequency and heavy claim payouts for insurers.
• **2. Moral Hazard and Fraudulent/Inflated Repair Claims:** Staged accidents, exaggerated garage repair bills, replacing old

parts with new parts under accident claims, and collusion between workshops and vehicle owners inflate insurer costs.
• **3. Inadequate Third-Party Premium Rates & Tariff Rigidities:** Statutory third-party premiums and fixed motor tariffs often fail to reflect individual driver risk history (lack of telematics or strict driver-record merit rating).
• **4. Disputes and Delays in Claim Settlement:** Conflicts over depreciation rates on spare parts, surveyor assessment delays, unlicensed driving, or drunken driving exclusions create dissatisfaction among policyholders.
• **5. Uninsured Vehicles and Hit-and-Run Cases:** Many two-wheelers and rural vehicles fail to renew their annual third-party insurance on time, leaving accident victims uncompensated in hit-and-run incidents.
• **6. Rising Cost of Imported Spare Parts and EV Battery Repairs:** With the rapid adoption of Electric Vehicles (EVs), expensive imported battery and sensor replacements have sharply increased average claim severity.

### Q17. Describe the process of calculating life insurance premium with suitable example.

**Answer (Process + Step-by-Step Numerical Example):**
**1. Three Core Actuarial Elements of Life Insurance Premium Calculation:**
Actuaries calculate life insurance premiums using three fundamental factors:
• **(a) Mortality Rate (q_x from Mortality Table):** The probability that a person of exact age *x* will die within the next year, derived from a standard Mortality Table.
• **(b) Assumed Rate of Interest (Investment Return, i):** Because premiums are collected at the beginning of the year and death claims are paid at the end of the year, insurers discount future expected claims at a conservative interest rate *i*.
• **(c) Loading for Expenses, Contingencies, and Profit (Bonus):** Adding administrative expenses, agent commissions, taxes, and safety/bonus margins to the Net Premium to arrive at the **Gross (Office) Premium**.

**2. Step-by-Step Process with a Suitable Numerical Example:**
**Example:** Suppose an insurer issues a 1-year Term Life Insurance policy of **Rs 100,000** to a group of **10,000 people aged 35**. From the mortality table, out of 10,000 persons aged 35, **20 persons are expected to die** during the year (Mortality probability q_35 = 20 / 10,000 = **0.002**). The insurer earns **6% interest p.a.** on invested funds, and expense loading is **15% of Net Premium**.

• **Step 1: Calculate Total Expected Death Claims at Year-End:**

Total Claims = 20 expected deaths x Rs 100,000 = **Rs 2,000,000** (or Expected Claim per person = 0.002 x Rs 100,000 = Rs 200).

• **Step 2: Discount Expected Claims by the Interest Rate to Find Net Single Premium (NSP):**

Present Value of Claims per policy (Net Single Premium) = Expected Claim / (1 + i)^1

NSP = Rs 200 / (1 + 0.06) = Rs 200 / 1.06 = **Rs 188.68 per policy**.

• **Step 3: Add Expense & Contingency Loading to Find Gross (Office) Premium:**

Gross Premium = Net Premium + Loading (15% of Rs 188.68)

Gross Premium = Rs 188.68 + Rs 28.30 = **Rs 216.98 per year**.

*(For multi-year policies, the Present Value of all future years' expected death benefits is divided by the Present Value of a Life Annuity Due of Re 1 to get the Net Level Annual Premium.)*

***EXAM HINT TO REMEMBER:*** *Remember the 3 pillars of Life Insurance Rate Making: (1) Mortality Rate + (2) Interest Rate (Discounting) = Net Pure Premium; then Net Pure Premium + (3) Expense Loading = Gross (Office) Premium!*

## Group 'C': Long Answer Questions (Attempt any THREE — ALL Solved) [3 x 10 = 30]

### Q18. Describe the concept of risk management. Also discuss the process of risk management.

**Answer:**
**1. Concept of Risk Management:**
Risk management is a proactive, scientific managerial process of identifying, analyzing, and evaluating pure (and enterprise) loss exposures faced by an organization or family, and selecting, implementing, and monitoring the most appropriate combination of risk control and risk financing techniques to minimize the adverse financial impact of losses and achieve pre-loss and post-loss objectives (survival, continuity of operations, earnings stability, and growth).

**2. Step-by-Step Process of Risk Management:**
• **Step 1: Establish Risk Management Objectives:**
Define clear *Pre-Loss Objectives* (economy, reduction of anxiety, meeting legal/statutory obligations) and *Post-Loss Objectives* (survival of the firm, continuity of operations, stability of earnings, continued growth, and social responsibility).
• **Step 2: Identify Loss Exposures (Risk Identification):**
Systematically uncover all property, liability, business interruption, human resource (death/disability/health), crime, and cyber exposures using risk analysis questionnaires, physical plant inspections, flowcharts, financial statements, and past loss histories.

• **Step 3: Measure and Analyze Loss Exposures (Risk Evaluation):**
Estimate the **Loss Frequency** (how often a loss is likely to occur) and **Loss Severity** (Maximum Possible Loss and Probable Maximum Loss) for each exposure, categorizing them into a 4-cell Frequency-Severity Matrix.
• **Step 4: Select the Appropriate Combination of Risk Treatment Techniques:**
- **(A) Risk Control Techniques (aimed at reducing frequency and severity):**
1. *Avoidance:* Abandoning or refusing to enter a high-risk activity (100% elimination of frequency).
2. *Loss Prevention:* Measures that reduce the **frequency** of losses (e.g., safety training, non-slip floors, cybersecurity firewalls).
3. *Loss Reduction:* Measures that reduce the **severity** of a loss once it occurs (e.g., automatic fire sprinklers, emergency medical response, off-site cloud backups, duplication and separation of inventory).
- **(B) Risk Financing Techniques (arranging funds to pay for losses that occur):**
1. *Retention (Self-Insurance / Deductibles):* Retaining High-Frequency / Low-Severity predictable losses out of working capital, funded reserves, or a captive insurer.
2. *Non-Insurance Transfer:* Shifting financial responsibility via hold-harmless contracts, leases, surety bonds, or derivative hedging.
3. *Commercial Insurance:* Transferring Low-Frequency / High-Severity catastrophic risks to professional insurers.
• **Step 5: Implement the Risk Management Program:**
Establish a formal Risk Management Policy Statement, coordinate with all functional departments, purchase selected insurance coverages, and install safety systems.
• **Step 6: Monitor, Review, and Evaluate the Program:**
Continuously audit loss experience via a Risk Management Information System (RMIS) and adjust coverage as business operations evolve.

### Q19. Insurance is a contract based on certain fundamental legal principles. Explain the principles of insurance contract.

**Answer:**
An insurance policy is a special legal contract governed by general contract law (offer and acceptance, consideration, legal capacity, and lawful purpose) plus **six fundamental legal principles** unique to insurance:

• **1. Principle of Utmost Good Faith (Uberrimae Fidei):**
Insurance contracts require a higher degree of honesty from both parties than ordinary commercial contracts. The applicant must voluntarily and truthfully disclose all material facts that would influence an underwriter's decision regarding premium rate or acceptance of risk. It is enforced through three legal doctrines:
- *Representations:* Statements made by the applicant on the proposal form (if material and false, the contract is voidable).
- *Concealment:* Intentional silence or failure to disclose a known material fact.
- *Warranties:* Strict conditions promised in the policy that must be literally complied with.

• **2. Principle of Insurable Interest:**
The insured must have a legally recognized pecuniary (financial) stake in the subject matter of insurance such that they benefit from its safety and suffer a direct financial loss upon its destruction, damage, or death.
- In *Life Insurance*, insurable interest must exist at the **inception** of the policy.
- In *Property & Marine Insurance*, insurable interest must exist at the **time of loss**.
- *Purpose:* Prevents gambling on strangers' lives/property and eliminates moral hazard.

• **3. Principle of Indemnity:**
States that the insurer will compensate the insured only up to the actual financial loss suffered (never exceeding the sum insured), restoring the insured to the same financial position they occupied immediately prior to the loss. It prevents the insured from profiting from a disaster. Actual Cash Value (ACV = Replacement Cost - Depreciation) is the primary measure of indemnity. *(Note: Life and personal accident policies are valued benefit contracts, not contracts of strict indemnity.)*

• **4. Principle of Subrogation:**
A corollary of indemnity: once the insurer pays a claim for a loss caused by the negligence or tort of a third party, the insurer steps into the legal shoes of the insured and acquires all legal rights to sue the negligent third party to recover the amount paid. This prevents the insured from collecting twice for the same loss.

• **5. Principle of Contribution:**
A second corollary of indemnity: when an insured holds two or more property/liability policies covering the same interest, same subject matter, and same peril (double insurance), they cannot recover more than the total actual loss. Each insurer pays only its proportionate (pro-rata) share of the loss.

- **6. Principle of Proximate Cause (Causa Proxima):**

When a loss results from a series of events, the insurer is liable only if the nearest, dominant, and efficient cause that set the chain of events in motion is an **insured peril** (and not an excluded peril).

### Q20. Explain the role of insurance industry in financial development of Nepal.

**Answer:**

The insurance industry in Nepal—comprising life insurers, non-life insurers, micro-insurers, and domestic reinsurance companies regulated by the Nepal Insurance Authority (NIA)—plays a vital catalyst role in the financial and economic development of the country:

- **1. Mobilization of Long-Term Domestic Savings & Capital Formation:**

Life insurance policies (endowment, whole life, child plans) collect small, regular contractual savings from millions of Nepalese households across urban and rural areas, turning fragmented household savings into a massive pool of long-term investable capital.

- **2. Major Institutional Investor in Banks and Infrastructure:**

Nepalese insurance companies are the largest institutional depositors in commercial and development banks (providing stable fixed deposits that strengthen banking sector liquidity) and invest heavily in government bonds, hydropower projects, agriculture, tourism infrastructure, and capital market securities (NEPSE).

- **3. Promoting Trade, Industry, and Banking Credit (Collateral Protection):**

Banks in Nepal require property, fire, marine, or motor insurance before sanctioning home loans, auto loans, industrial project loans, or import Letters of Credit. By absorbing asset destruction risk, insurance enables the banking system to expand credit safely.

- **4. Protecting Rural Livelihoods through Agriculture and Micro-Insurance:**

Subsidized crop and livestock insurance and dedicated micro-insurance companies protect low-income farmers and micro-entrepreneurs against natural disasters, preventing rural families from falling back into poverty.

- **5. Retaining Foreign Exchange via Domestic Reinsurance:**

The establishment of domestic reinsurers (Nepal Reinsurance Company and Himalayan Reinsurance) retains billions of Rupees of reinsurance premiums within Nepal's financial system and earns foreign currency by accepting inward reinsurance from abroad.

- **6. Employment Generation, Tax Revenue, and Social Security:**

Creates direct and indirect employment for thousands of employees, surveyors, and over 200,000 licensed agents, contributes substantial corporate and withholding taxes to the government, and reduces the state's post-disaster fiscal burden.

### Q21. Regulators regulate various aspects of the insurance company operation. In light of this statement, explain the areas of insurance regulation.

**Answer:**

Because insurance companies act as fiduciaries managing public funds to meet contingent future promises, insurance regulators (such as the Nepal Insurance Authority under the Insurance Act 2079) supervise virtually every major area of an insurer's operations:

- **1. Licensing and Formation of Insurers (Entry Regulation):**

Setting minimum paid-up capital requirements (e.g., Rs 5 billion for life insurers and Rs 2.5 billion for non-life insurers in Nepal), vetting promoter fit-and-proper criteria, and issuing operating licenses for life, non-life, micro-insurance, and reinsurance businesses.

- **2. Financial Solvency, Capital Adequacy, and Reserve Regulation:**

Mandating strict Risk-Based Capital (RBC) and solvency margin ratios, unearned premium reserves, statutory catastrophe reserves, and actuarial valuation of life funds to prevent insurer insolvency.

- **3. Regulation of Investment of Insurance Funds:**

Issuing investment directives that specify minimum and maximum percentages of technical reserves that insurers may invest in government bonds, BFIs' fixed deposits, listed shares, debentures, real estate, and infrastructure/hydropower projects—ensuring safety, liquidity, and diversification.

- **4. Rate Making and Premium Tariff Regulation:**

Ensuring that premium rates are adequate (to maintain solvency), not excessive (to protect consumers), and not unfairly discriminatory, including setting tariffs for compulsory third-party motor, property/fire, and subsidized agriculture insurance.

- **5. Policy Contract Forms and Product Approval:**

Reviewing and approving all new life and non-life insurance policy wordings, endorsements, and actuarial product filings before they can be sold to the public so unfair exclusions or deceptive clauses are eliminated.

• **6. Regulation of Market Conduct, Intermediaries, and Corporate Governance:**

Licensing and training insurance agents, brokers, and loss surveyors; enforcing anti-money laundering (AML/CFT) rules; capping CEO/board governance limits; and prohibiting rebating or misrepresentation.

• **7. Claim Settlement Supervision and Policyholder Grievance Redressal:**

Enforcing mandatory timelines for surveyor reports and claim payments, and acting as a quasi-judicial body to resolve disputes between policyholders and insurers.

## Group 'D': Comprehensive Answer / Case / Situation Analysis [20 Marks]

**Q22. Case Study: Bikash Koirala & Horizon Digital Advisory Bikash Koirala (39-year-old former Operations Director, sole financial provider for his spouse and two school-going children, active stock investor and short-term trader) resigned from his corporate job to launch Horizon Digital Advisory in Lalitpur (IT and digital consulting for microfinance institutions across Nepal, funded with personal savings, 6 employees, servers/networking devices/premium laptops, home and vehicle loans). a. Define pure risk and speculative risk. Also identify all the pure risks and speculative risks faced by Mr. Koirala and his organization. b. Describe the type of life insurance policy that is suitable for Mr. Koirala. What additional benefits could be added to the policy to enhance the coverage? c. Suggest an insurance policy to cover the risk of fire and related perils. d. Why do you think is the decision to provide Group Health Insurance for the staffs considered as a risk management tool, rather than just an additional expense to the organization?**

**Answer (Comprehensive Case Study Solution):**

**a. Definitions and Identification of Pure Risks and Speculative Risks Faced by Mr. Koirala and Horizon Digital Advisory:**

• **Definition of Pure Risk:** A category of risk in which there are only two possible outcomes—**loss or no loss** (zero possibility of financial gain). Pure risks are generally accidental, unintentional, and **insurable** (e.g., fire, earthquake, premature death).

• **Definition of Speculative Risk:** A category of risk in which there are three possible outcomes—**loss, no loss (breakeven), or financial gain/profit**. Speculative risks are undertaken voluntarily in pursuit of profit and are generally not insurable through traditional insurance (handled instead via diversification and hedging).

• **Pure Risks Faced by Mr. Koirala and Horizon Digital Advisory:**

1. *Personal / Family Pure Risks:* Untimely (premature) death of Mr. Koirala, permanent disability, or diagnosis of a critical illness—which would cut off his primary income and leave his family unable to pay daily living costs, two children's schooling, and ongoing home and vehicle loans.

2. *Property Pure Risks:* Physical damage or destruction of the Lalitpur office premises, servers, networking devices, and premium laptops due to **fire, electrical damage, theft, or natural disasters like earthquakes**.

3. *Liability & Operational Pure Risks:* Breaches of sensitive client financial data or system failures caused by human error, omissions, or negligence during IT consulting delivery—exposing the firm to expensive client compensation claims and legal defense costs; employee sickness or medical emergencies among his six staff members.

• **Speculative Risks Faced by Mr. Koirala and His Organization:**

1. *Stock Market Investment & Short-Term Trading Risk:* Price fluctuations in Mr. Koirala's equity investments and short-term trading positions (can result in capital gains or trading losses).

2. *Entrepreneurial Startup Venture Risk:* Committing a large share of his personal savings to launch Horizon Digital Advisory—the firm may prosper in the microfinance IT consulting market or incur operating losses due to market competition.

**b. Suitable Life Insurance Policy for Mr. Koirala and Additional Benefits (Riders) to Enhance Coverage:**

• **Recommended Life Insurance Policy:** A high-coverage **Term Life Insurance Policy** (or a combination of a 20–25 year Term Life Policy plus an Endowment/Whole Life base policy) sized to cover his home loan, vehicle loan, children's education fund, and family living expenses.

- *Why Suitable:* Since Mr. Koirala has already committed a large share of his personal savings into launching his startup, a Term Life policy gives his family maximum lump-sum financial protection at the most affordable premium, keeping cash available for Horizon Digital Advisory.

• **Additional Benefits (Riders) to Add to the Policy:**

1. *Critical Illness (CI) Benefit Rider:* Pays an immediate tax-free lump sum upon diagnosis of major critical illnesses (cancer, heart attack, stroke, kidney failure, organ transplant) to pay for specialized medical care and replace lost income during recovery.

2. *Permanent Total Disability (PTD) & Accidental Death and Dismemberment (AD&D) Rider:* Provides income replacement / lump-sum payout if an accident or illness leaves him permanently unable to work.

3. *Waiver of Premium Rider:* Automatically waives all future policy premiums if Mr. Koirala becomes permanently disabled, ensuring the life insurance policy remains active without financial burden.

**c. Suggested Insurance Policy to Cover the Risk of Fire and Related Perils (Office & IT Equipment):**
Mr. Koirala should purchase a **Standard Fire and Allied Perils Insurance Policy (Commercial Property / Office Package Policy)** coupled with an **Electronic Equipment Insurance (EEI) / Burglary Endorsement**:
• **1. Standard Fire and Allied Perils Coverage:** Indemnifies the Lalitpur office premises, furniture, and fixtures against direct physical loss from fire, lightning, explosion, and electrical short-circuit damage, extended with allied perils endorsements for **Earthquake, Flood, Inundation, Landslide, Storm, and Riot/Strike/Malicious Damage (RSMD)**.
• **2. Electronic Equipment Insurance (EEI) & Burglary/Theft Coverage:** Specifically covers high-value IT infrastructure—servers, networking switches, and premium laptops—against internal electrical/mechanical breakdown, power surges, and forcible theft/burglary.
• **3. Business Interruption (Consequential Loss) & Professional Indemnity Add-on:** Covers lost revenue during post-fire recovery alongside a Professional Indemnity (E&O) policy for client data-breach liabilities.

**d. Why Providing Group Health Insurance for Staff is a Risk Management Tool Rather Than Just an Additional Expense:**
• **1. Controlling Human Resource & Operational Continuity Risk:** In a specialized 6-person IT consulting team serving microfinance institutions, losing even one key engineer to prolonged untreated illness disrupts client projects. Prompt access to quality healthcare ensures faster recovery and minimizes absenteeism.
• **2. Talent Retention and Reduction of Turnover Risk:** Young startups face high risk of losing trained IT staff to larger corporations. Group health insurance builds employee loyalty, job satisfaction, and retention, avoiding the high cost of recruiting and retraining replacements.
• **3. Reducing Attitudinal Hazard and Human Error:** Employees who are financially secure regarding their family's medical needs experience lower stress and higher concentration, directly reducing the human errors that cause system failures or client data breaches.
• **4. Predictable Budgeting & Tax Efficiency:** Converts unpredictable employee medical emergency requests into a small, predictable, tax-deductible annual premium with pooled group underwriting benefits.